

1 Edits & required changes

3 • In the end, modify extend supplementary & figure summary section

- 4 • Fix all ρ according to the table below. Also see specific figure changes.
 - 5 – Figure 1: 2nd panel, call it ρ^{fit} ; third panel call it ρ_{NK} ; also third panel add b to base chance
 - 6 and s to splitting (like in fourth panel)
 - 7 – Figure 3: Everywhere, check that maths is italic, e.g. in **A** Generations M (check space)
 - 8 and Fitness F_{μ} .
 - 9 * A: use ρ_{NK} in legend instead of $(K + 1)/N$
 - 10 * B: Use ρ_{NK} instead of $(K + 1)/N$ (both legend and axes); use Mean ρ^{fit} instead of
 - 11 Mean estimated ρ (both legend and axes)
 - 12 * C: Use ρ_{NK} instead of $(K + 1)/N$ (both legend and axes); use Mean ρ^{fit} instead of
 - 13 Mean estimated ρ (both legend and axes)
 - 14 * D: Use ρ_{NK} instead of $(K + 1)/N$ (both legend and axes); use Mean ρ^{fit} instead of
 - 15 Mean estimated ρ (both legend and axes)
 - 16 * E: Use ρ_{NK} instead of $(K + 1)/N$ in axis
 - 17 – Figure 4: Everywhere, check that maths is italic, e.g. in **A** Frequency index i or Power
 - 18 spectral coefficients b_i .
 - 19 * B: ρ_2^{loc} vs ρ_2^{glob} in title; ρ_2^{loc} Y axis; ρ_2^{glob} in legend; ρ_{NK} in X axis
 - 20 * C: ρ_2^{loc} variability in title
 - 21 * D: ρ_2^{fit} in title and Y axis
 - 22 * E: ρ_2^{loc} or ρ_2^{glob} in title and Y axis (depending on whether sampled from one or more
 - 23 starting points)
 - 24 * F: ρ_2^{loc} or ρ_2^{glob} in title and Y axis (depending on whether sampled from one or more
 - 25 starting points)
 - 26 * G: First panel, call it ρ_2^{loc} or ρ_2^{glob} (depending on whether sampled from one or more
 - 27 starting points; I think multiple); second panel call it ρ_2
 - 28 – Figure 5: Everywhere, check that maths is italic, e.g. in **A** Predicted base chance b (check
 - 29 space).
 - 30 * A: Replace ρ_{NK} in title and X axis; “total 1200” is a bit strange in right Y axis—can this
 - 31 be moved to the caption?
 - 32 * D–G: use ρ_2 ; label decay (Y axis) with G_{μ}
 - 33 – Figure 6: Call it ρ_2^{fit} everywhere; for d–f use same markers as in other plots
 - 34 – Figure 7: in B and C, call it ρ_2
 - 35 – Figure S2: label fitness decay with either F_{μ} or G_{μ} (I guess the latter)

- 36 – Figure S3: use the correct rho depending on mutation model. **This figure is currently**
- 37 **not cited in text—please reference somewhere!!!**
- 38 – Figure S5: replace true rho with ρ_{NK}

Table 1. Summary of revised ruggedness and decay-rate notation.

Old var	New var	Description
ρ	ρ	Analytical decay-rate metric for the mean fitness decay F_μ under the mutation model.
ρ or ρ_Δ	ρ_2	Analytical squared-decay metric for $G_\mu = \mathbb{E}_v[F_\mu^2]$; corresponds to the normalised Dirichlet-energy expression.
ρ	ρ^{fit}	Fitted decay rate estimated from finite/noisy observations of F_μ .
ρ	ρ_2^{fit}	Fitted squared-decay rate estimated from finite/noisy observations of G_μ .
ρ	ρ^{loc}	Local fitted estimate from F_μ using one or a small number of starting genotypes.
ρ	ρ_2^{loc}	Local fitted estimate from G_μ using one or a small number of starting genotypes.
ρ	ρ^{glob}	Global fitted estimate from F_μ after averaging over all, or a representative set of, starting genotypes.
ρ	ρ_2^{glob}	Global fitted estimate from G_μ after averaging over all, or a representative set of, starting genotypes.
$(K + 1)/N$ or ρ	ρ_{NK}	Closed-form NK ruggedness value, defined as $\rho_{NK} \stackrel{\text{def}}{=} (K + 1)/N$; for NK landscapes, $\rho \approx \rho_{NK}$.
ρ	ρ_{HoC}	Closed-form house-of-cards ruggedness value; for zero-mean HoC landscapes, $\rho_{\text{HoC}} = 1$.
ρ	$\tilde{\rho}_2$	Analytical squared-decay metric for G_μ under a weighted, undirected biased mutation operator.
ρ or $\bar{\rho}_\Delta$	$\bar{\rho}_2$	Analytical squared-decay metric for G_μ under a directed or asymmetric mutation operator, using the symmetrised Laplacian approximation.

Sequence-free landscape inference for directed evolution

Sebastian Towers¹, Jessica James¹, Harrison Steel¹, Idris Kempf¹✉

¹Department of Engineering Science, University of Oxford, Oxford, United Kingdom

✉ For correspondence:

idris.kempf@eng.ox.ac.uk, harrison.steel@eng.ox.ac.uk

Equal contribution: [†]These authors contributed equally to this work.

Data Availability Statement:

Data and code used for running experiments and plotting is available on Github at <https://github.com/jessmja/SLIDE>. Raw data available at [doi:10.5281/zenodo.16849761](https://doi.org/10.5281/zenodo.16849761).

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Abstract

Directed evolution is a method for engineering biological systems or components, such as proteins, wherein desired traits are optimised through iterative rounds of mutagenesis and selection of fit variants. The process of protein directed evolution can be envisaged as navigation over high-dimensional optimisation landscapes with numerous local maxima. The performance of any strategy in navigating such a landscape is dependent on the ruggedness of that landscape. However, this information is generally unavailable at the outset of an experiment. Here we propose **SLIDE**, Sequence-free Landscape Inference for Directed Evolution, which consists of two parts. First, SLIDE provides an estimation for landscape ruggedness from a mutating population using only population-level phenotypic data and an estimation of mutation rate. Such ruggedness information in itself is valuable in protein design, for instance in predicting evolutionary stability. Second, SLIDE offers a framework for using the estimated ruggedness metric to identify high-performing selection strategies for directed evolution. Using theoretical NK landscapes and four empirical protein fitness landscapes, we demonstrate *in silico* consistent improvement upon the performance of standard selection strategies, particularly on rugged landscapes, using a pipeline that could also be combined with emerging AI-based methods for driving directed evolution.

Author Summary

Directed evolution (DE) navigates high-dimensional protein fitness landscapes to find improved variants, but the success of any strategy depends on the landscape's ruggedness, information that is typically unknown and often requires genetic sequencing to derive. We present Sequence-free Landscape Inference for Directed Evolution (SLIDE), a practical two-step framework that (i) estimates landscape ruggedness from simple population-level phenotypic decay curves alongside mutation rate estimation, and (ii) uses that ruggedness estimate to choose DE control parameters

that better balance exploration and exploitation. In contrast to existing ruggedness metrics, SLIDE relies on spectral analysis to link the exponential decay rate of mean fitness under random, unbiased mutations to the dominant frequency content of the landscape. This decay rate is related to the normalised Dirichlet energy of the underlying genotype graph, and the framework naturally extends to biased mutation spectra. We combine SLIDE with DE strategies developed in our previous work, but the method is strategy-agnostic and can be paired with any DE workflow, including emerging AI-driven workflows. *In silico* tests show that SLIDE reliably recovers ruggedness and improves search efficiency (for example, up to 44.2% relative fitness gain versus a common baseline on a TEV protease dataset). Because SLIDE does not require extensive sequencing beyond a mutation rate estimation, it lowers experimental cost and scales to large screens. Beyond DE, the approach can rapidly characterise landscapes arising in genetic circuits, enzyme libraries, and ecological models, providing a way to quantify genetic stability and guide experimental or computational search.

Introduction

Fitness landscapes map DNA or amino acid sequences to a measure of fitness such as enzyme activity, cell growth, binding affinity, or stability [1]. Fitness landscapes are characterised by multiple structural properties, such as ruggedness, epistasis, neutrality, and navigability, which have significant consequences for evolution and stability [2, 3]. In this work, we focus on inferring the ruggedness of a fitness landscape, which unlocks the ability to assess the robustness of a protein to mutations and to optimise directed evolution (DE) protocols.

Various metrics have been proposed to define and quantify the ruggedness of fitness landscapes [4–8]. Although these metrics have been shown to correlate across several empirical landscapes [8], most of them require combinatorially-complete phenotype-genotype maps to compute a ruggedness measure. Several of these metrics come from spectral landscape theory [9], which describes fitness landscapes using tools of Fourier analysis. In the same way that a signal can be decomposed into a sum of waves, a fitness landscape can be decomposed into a sum of landscapes of different frequencies (ruggedness). Where low frequency terms dominate, the landscape can appear smooth, whereas where high frequency terms dominate, the landscape can appear rugged.

In this paper, we develop Sequence-Free Landscape Inference for Directed Evolution (SLIDE) to efficiently estimate the ruggedness of a landscape using phenotypic information alongside an estimated mutation rate (Fig 1), and we validate this pipeline *in silico*. Based on spectral landscape theory and under the assumption of random, unbiased mutations, we define a novel ruggedness metric as a normalised Dirichlet energy of the genotype graph, which can be related to the dominant Fourier terms of the landscape. We show that this metric can be estimated from the average fitness decay of a population that is randomly mutated from a starting point, contrasting existing ruggedness metrics that require combinatorially-complete phenotype-genotype maps. The

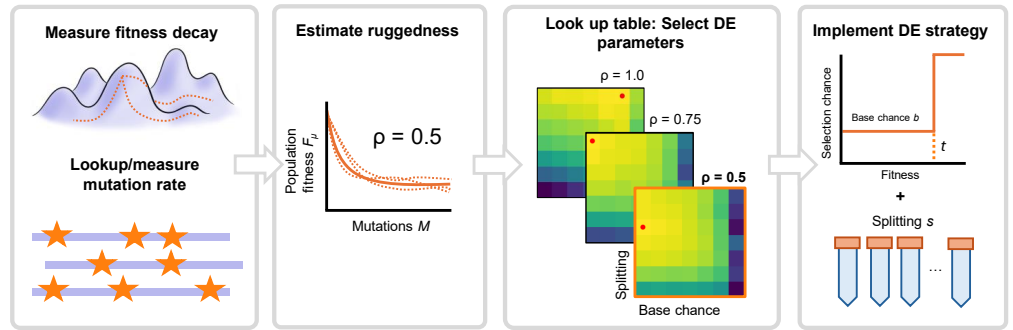


Figure 1. Schematic outline of a future DE pipeline. The initial phase of SLIDE involves random mutation away from one or several starting points, and measurement of the resulting fitness decay curve, which alongside a known mutation rate, can be used to estimate the fitted decay rate ρ^{fit} as a measure of ruggedness. A look-up table matches analytical ρ values of NK landscapes to optimal DE strategy parameters. This look-up table can be used to select appropriate parameters for the measured ρ^{fit} . Parameters explored in this paper include base chance (applying a constant chance of selection that is irrespective of fitness) and population splitting [10].

104 ruggedness metric characterises the rate at which the mean population fitness relaxes toward
 105 the landscape average; on smooth landscapes, this average fitness decays more slowly than on
 106 rugged landscapes. This phenomenon can be attributed to fitness correlations between variants
 107 on smooth landscapes, as random mutations are less likely to result in large fitness differences.
 108 Most importantly, the fitness decay curve can be acquired without large-scale sequencing data;
 109 instead it uses population-level fitness readouts and a mutation rate estimation.

110 In SLIDE, we combine ruggedness inference with DE. In DE, biological components such as proteins
 111 are engineered and improved through iterative rounds of selection and mutagenesis [11]. It can
 112 be seen as navigating a high-dimensional fitness landscape, aiming to guide the fitness of variants
 113 to a high-performing location on the landscape. Early DE optimisation techniques such as ProSAR
 114 used statistical models to correlate sequence with measured activity and iteratively design vari-
 115 ants that maximize predicted fitness [12]. Since then, advances in machine learning have led to
 116 the development of more sophisticated algorithms that actively design the sequence of variants
 117 that comprise the next generation [13–15]. Although effective, these methods are limited by their
 118 need for high-throughput sequencing data. Without sequencing data, most DE experiments simply
 119 select the top proportion of variants in each iteration [16–18], i.e. truncation selection (comparable
 120 to hill-climbing algorithms), which we refer to as the "baseline strategy". This approach results in
 121 minimal landscape exploration, and is hence prone to becoming trapped in local optima [10].

122 To address this limitation, several methods have been proposed for balancing landscape explo-
 123 ration and exploitation. Such methods include applying non-constant selection pressure [19], in-
 124 creasing the proportion of population selected [20], splitting the population into sub-populations,
 125 and including a base chance of selection that is irrespective of fitness [10]. However, effective appli-
 126 cation of these methods depends on knowledge of landscape properties such as ruggedness [10].

127 Inferring the ruggedness beforehand is essential for selecting strategy parameters to ensure the
 128 balance of exploration and exploitation. Here, we select these parameters based on a ruggedness
 129 estimation through SLIDE, where ruggedness and strategy parameters are identified prior to DE
 130 experiments. By doing so, SLIDE increases the likelihood of reaching higher-fitness variants within
 131 a fixed number of generations, even when the decay rate is not estimated accurately.

132 Any method for DE—both with and without sequencing—can integrate SLIDE to tune DE strategy
 133 parameters. In this paper, SLIDE is interfaced with strategies developed in our previous work [10],
 134 controlling base chance and population-splitting parameters. We evaluate our method for DE *in*
 135 *silico* on theoretical landscapes obtained from the NK model [21] and on four empirical land-
 136 scapes [2, 22–24]. We show that SLIDE consistently finds the optimal parameters on NK land-
 137 scapes, and outperforms or matches the baseline strategy on all empirical landscapes.

138 Results

139 Ruggedness inference from fitness decay rate

140 Ruggedness of a fitness landscape can be defined in terms of various mathematically distinct met-
 141 rics (see [4–8]), which aim to capture how fitness changes when the landscape is traversed. Fig 2a-b
 142 present examples of smooth and rugged landscapes for a protein of length $N = 2$ with 16 geno-
 143 types, where σ_i is used to denote the allele at gene locus $i = \{1, 2\}$ (i.e. which is a choice from A
 144 possible amino acids, here $A = 4$). Intuitively, smooth landscapes should have high fitness variants
 145 clustered together, leading to high fitness correlations and relatively few local extrema. In contrast,
 146 rugged landscapes should exhibit large fitness differences between adjacent nodes, resulting in
 147 more local extrema and lower fitness correlations.

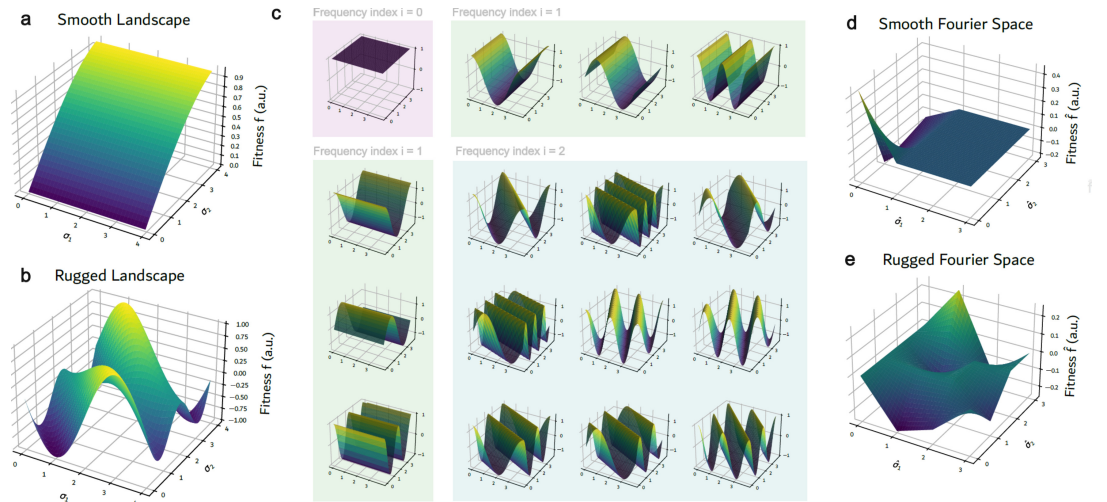


Figure 2. Demonstration of landscape transformations. Examples of **A:** smooth and **B:** rugged landscapes. **C:** Example Fourier basis. **D–E:** Smooth and rugged landscapes transformed into Fourier space with the example basis.

148 We assume that the landscape is traversed via unbiased, single-point mutations and develop a
 149 method to measure ruggedness based on spectral landscape theory [9], which decomposes land-
 150 scapes into low-frequency (smooth) and high-frequency (rugged) components using a Graph Fourier
 151 transform (GFT). Analogous to a standard Fourier transform, the GFT decomposes the original land-
 152 scape f to express it as a mixture of basis landscapes of increasing frequency, such as those shown
 153 in Fig 2c. Here, f can be interpreted as mapping from genotype space $\mathcal{G}$ to fitness, $f : \mathcal{G} \rightarrow \mathbb{R}$, or
 154 as a vector $f \in \mathbb{R}^{|\mathcal{G}|}$, where $|\mathcal{G}|$ denotes the cardinality of $\mathcal{G}$. For a protein of length N and A choices
 155 of amino acids, there are $|\mathcal{G}| = A^N$ basis landscapes grouped into $N + 1$ frequencies $\lambda_0, \dots, \lambda_N$. The
 156 weights of each basis landscape in f can be arranged to form a new landscape $\tilde{f}$ in Fourier do-
 157 main, such as shown in Fig 2d-e. In the Fourier domain, the alleles σ_i are replaced with the Fourier
 158 components $\hat{\sigma}_i$ and the genotype fitness with the weight of the Fourier component $\hat{\sigma}_i$.

159 Consider a population $\mathbf{v}$ subjected to random unbiased mutations. Let F_μ denote the expected
 160 population fitness after an average of μ accumulated mutations per individual, where the expecta-
 161 tion is taken over mutation trajectories. We further define the average squared fitness $G_\mu \triangleq \mathbb{E}_\mathbf{v}[F_\mu^2]$,
 162 where the expectation is additionally taken over all single-genotype populations. Based on the GFT,
 163 we propose to extract the scalar ruggedness metrics ρ and ρ_2 , associated with F_μ and G_μ , respec-
 164 tively, as the weighted frequency averages

$$\rho = \frac{1}{d} \frac{\sum_{i=0}^N a_i \lambda_i}{\sum_{i=0}^N a_i - a_0} = \frac{1}{d} \frac{\mathbf{f}^\top \Delta \mathbf{v}}{\mathbf{f}^\top \mathbf{v} - \bar{f}}, \quad (1a)$$

$$\rho_2 = \frac{1}{d} \frac{\sum_{i=0}^N b_i \lambda_i}{\sum_{i=0}^N b_i - b_0} = \frac{1}{d} \frac{\mathbf{f}^\top \Delta \mathbf{f}}{\mathbf{f}^\top \mathbf{f} - A^N \bar{f}^2}. \quad (1b)$$

165 where $d = N(A - 1)$ is a factor determined by landscape size/dimension, a_i the population-weighted
 166 spectral coefficient at frequency index, b_i the population-independent power spectral coefficient
 167 at frequency index i , $\bar{f}$ the landscape average, and λ_i a frequency-specific parameter that depends
 168 on the genotype graph Laplacian Δ resulting from the unbiased mutation process (see Methods).
 169 The quantity in the first part of Eq (1b) represents the centre of mass of the spectrum. The second
 170 part shows that this spectral centroid is equivalent to a normalised Dirichlet energy of the under-
 171 lying genotype graph, which quantifies the average squared fitness variation across mutational
 172 neighbours. As such, the frequency domain interpretation of our metrics is naturally tied to the
 173 unbiased mutation model, though in Methods we show how it extends to biased mutation opera-
 174 tors beyond the GFT. The metrics from Eq (1) are unique because they can be easily computed from
 175 phenotypical data extracted from a randomly mutagenised population. In Methods, we show that
 176 the expected population fitness F_μ , the average squared fitness G_μ , and the metrics from Eq (1) are

177 related by:

$$F_\mu \approx C e^{-\mu\rho} + \bar{f}, \quad (2a)$$

$$G_\mu \approx C_2 e^{-2\mu\rho_2} + \bar{f}^2, \quad (2b)$$

178 where C is a constant that depends on the initial population and can be negative if the initial pop-
 179 ulation fitness is lower than $\bar{f}$, and C_2 is a population-independent constant with $C_2 \geq 0$ if G_μ is
 180 obtained from *all* single-genotype populations. The ruggedness metrics ρ and ρ_2 therefore govern
 181 the rate at which the mean population fitness and its square relax toward the landscape average
 182 $\bar{f}$ and $\bar{f}^2$. In both cases, the decay rates can be interpreted as the ruggedness of the landscape
 183 by relating it to the different Fourier components from Fig 2: for large ρ or ρ_2 , higher-frequency
 184 terms dominate the corresponding decay, and for small ρ or ρ_2 , lower-frequency terms dominate.
 185 On more rugged landscapes, the decay rate ρ_2 will be larger than on smoother landscapes, and the
 186 mean population fitness will converge more quickly to the landscape average. This observation is
 187 the essential phenomenon that our work exploits to estimate the landscape ruggedness through
 188 measurements of the mean fitness of a population as it accumulates mutations. In other words,
 189 suppose that each population member is mutagenised with θ single-point mutation per genera-
 190 tion on average, and the mean fitness recorded as $F_{\mu_1}, F_{\mu_2}, \dots, F_{\mu_M}$ with $\mu_i = \theta i$ over M generations.
 191 These values can be used to estimate the parameters C , ρ , and $\bar{f}$ in Eq (2a). Equivalently, after
 192 squaring these values and averaging over starting populations, they can be used to estimate C_2 , ρ_2 ,
 193 and $\bar{f}^2$ in Eq (2b).

194 In the following, we denote decay rates fitted from finite, noisy observations by ρ^{fit} for F_μ and ρ_2^{fit} for
 195 G_μ . The metric ρ is associated with the fitness decay from a particular starting genotype or popula-
 196 tion and therefore represents a “local” or a starting-population-dependent notion of ruggedness.
 197 By contrast, ρ_2 is associated with G_μ , which averages over all starting genotypes, and therefore
 198 represents a global property of the landscape. We further distinguish between local estimates ρ_2^{loc} ,
 199 obtained from one or a small number of starting genotypes, and global estimates ρ_2^{glob} , obtained
 200 after averaging over all, or a large representative set of, starting genotypes. The analytical quan-
 201 tities ρ and ρ_2 correspond to spectral averages of the underlying landscape, whereas the fitted
 202 quantities are obtained by approximating finite observations with a single exponential decay. In
 203 homogeneous landscapes, such as NK landscapes, local and global notions of ruggedness broadly
 204 coincide. In particular, the epistatic interaction parameter K defines the closed-form ruggedness
 205 value $\rho_{NK} \stackrel{\text{def}}{=} (K+1)/N$, and the analytical decay rates satisfy $\rho \approx \rho_2 \approx \rho_{NK}$ (see Eq. (S2) in Supporting
 206 Information Section S4). Consequently, the fitted and analytical decay rates also broadly coincide,
 207 i.e. $\rho^{\text{fit}} \approx \rho$ and $\rho_2^{\text{fit}} \approx \rho_2^{\text{loc}} \approx \rho_2^{\text{global}} \approx \rho_2$. Beyond NK landscapes, these metrics broadly coincide if the
 208 landscape is dominated by a single spectral frequency. In heterogeneous landscapes where more
 209 than one frequency dominate, however, local and global notions of ruggedness need not agree,

and fitted estimates may differ from the corresponding analytical values.

The trend from Eq (2a) is illustrated in Fig 3a, which shows the mean population fitness decay F_μ from mutating populations on two NK landscapes of differing ruggedness. Figure 3a shows that on landscapes with greater NK ruggedness ($\rho_{NK} = 0.75$), the rate of fitness decay is higher than that of the smooth landscape ($\rho_{NK} = 0.1$). This is further validated in Fig 3b, which shows that the fitted estimate ρ^{fit} correlates strongly with $\rho_{NK} \in \{0.1, 0.2, \dots, 1\}$. Moreover, as all estimates are computed from single starting genotypes, Figure 3b and the derivations from Supporting Information Section S4 show that on NK landscapes, the decay rates satisfy $\rho \approx \rho^{\text{fit}} \approx \rho^{\text{loc}} \approx \rho^{\text{glob}}$ and $\rho_2 \approx \rho_2^{\text{fit}} \approx \rho_2^{\text{loc}} \approx \rho_2^{\text{glob}}$, as well as $\rho \approx \rho_2$.

Due to the noise associated with observing a randomly mutating population, the reliability of ρ^{fit} estimates also depends on the parameters of the observed population, including the population

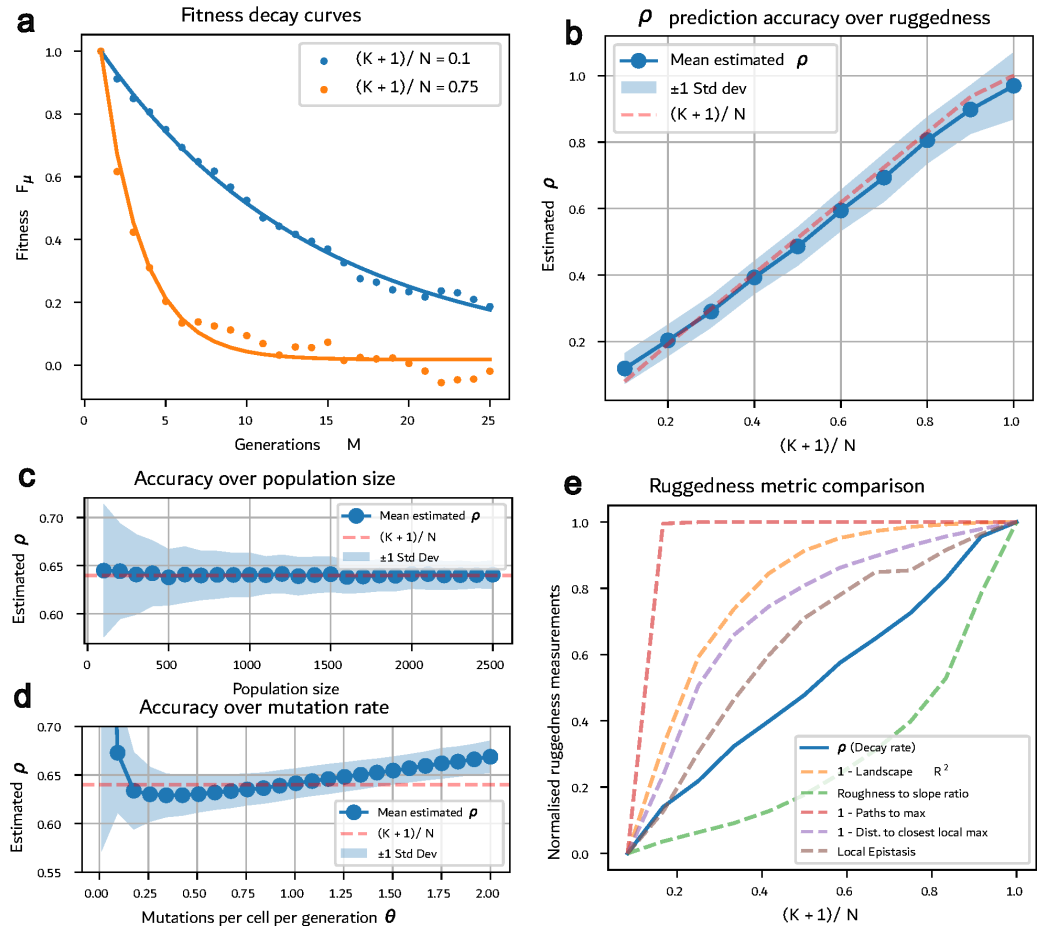


Figure 3. Fitness decay rate of a randomly mutating population, as a metric for landscape ruggedness. **A:** Simulations of two populations on two NK landscapes ($\rho_{NK} = 0.1$ and 0.75 , respectively, with $N = 20$) mutating away from a starting point (population size = 300, mutation rate $\theta = 0.5/N = 0.025$ per generation). Exponential fits to Eq (2a) are indicated by solid lines. **B:** Fitted ρ^{fit} values across landscapes of increasing ruggedness using $M = 25$ generations and a mutation rate of $\theta = 0.025$ per generation. The figure shows mean and standard deviation for 300 estimates. **C-D:** Accuracy of ρ^{fit} over increasing population size (C) and mutation rate θ (D) for 500 estimates. **E:** Comparison of the analytical decay rate ρ with other ruggedness metrics on 600 NK landscapes of increasing ρ_{NK} . All metrics are normalised between 0 (least rugged) and 1 (most rugged).

size and the mutation rate θ . Figure 3c shows ρ^{fit} estimates over increasing population sizes. For each population size, ρ^{fit} is estimated 500 times with Fig 3 showing mean and standard deviation (Std Dev). For small population sizes, the ruggedness estimate exhibits higher variability due to noise in the mean population fitness. For population sizes exceeding 2,000, the standard deviation is less than 5 % of the analytical decay rate. The effect of mutation rate is analysed in Fig 3d; for small θ , only a small fraction of the fitness decay curve in Fig 3a is captured, leading to an increased variability of the estimate from equation (2a). For very large θ , the mean population fitness decays more rapidly to the landscape average, again leading to a large variability of the estimate from Eq (2a) since only few datapoints characterise the decay. In practice, an estimate of the average mutation rate per generation is required, which is assumed to be constant across generations (see Eq (3) in Methods). If the mutation rate per generation θ is not exactly known but estimated as $\tilde{\theta}$, ρ^{fit} will scale with $\theta/\tilde{\theta}$ as long as the mutation rate is constant across generations, e.g. if the actual θ is twice as high as $\tilde{\theta}$ then ρ^{fit} will be twice as large. As such, if the mutation rate error is constant across analysed landscapes, fitted ρ^{fit} values remain comparable, i.e. the ordering resulting from the metric is preserved under an estimation error.

To further validate Eq (2a) and its characteristic decay rate ρ , Fig 3e provides a comparison between ρ and other ruggedness measures across NK landscapes (See Supporting Information Section S3). The other ruggedness measures are the landscape R^2 measure [8], the spectral entropy, the roughness to slope ratio [4], number of paths to the global maximum from the opposite position on the landscape [7], the distance to the nearest local maximum [6], and the local epistasis metric [8]. Figure 3e shows that although all metrics increase with ρ_{NK} , they express either concave (roughness to slope ratio), linear (decay rate), or convex curves (all other metrics). Because all metrics increase with K , this allows one to compare the ordering (w.r.t. ruggedness) of landscapes across metrics, but the differences in curve shapes prohibits comparison of distances (i.e. quantitative ruggedness differences) between landscapes across metrics. However, one can mathematically prove that the decay rate ρ directly correlates with ρ_{NK} (see Eq (S2) in the Supporting Information), and the same relationship holds regardless of N . Overall we believe this provides a second benefit of our methodology—that it provides an interpretable measure of ruggedness that has immediate meaning in the widely-studied NK landscapes.

Ruggedness inference on empirical landscapes

The performance of SLIDE on empirical landscapes was assessed *in silico* by testing it on four experimentally-measured landscapes: GB1 (4 sites) [2], ParD3 (3 sites) [22], TrpB (4 sites) [23] and TEV (4 sites) [24]. The Fourier spectra of each landscape are shown in Fig 4a, with the dashed vertical lines denoting the analytical ρ_2 from Eq (1b). Here, in contrast to NK landscapes, the heterogeneity of empirical landscapes or—in other words—the spread of spectra make it necessary to distinguish

between the “local” or starting population-dependent ρ and ρ^{fit} , and the global landscape property ρ_2 and ρ_2^{fit} . Figure 4b compares ρ^{fit} obtained from single-genotype starting populations (shaded) with the analytical ρ_2 obtained from Eq (1b). On NK landscapes, these values broadly correspond, indicating that NK landscapes are effectively homogeneous. On empirical landscapes, ρ^{fit} and ρ_2 do not correspond. However, the starting population-dependent estimates may still be used to measure the ruggedness of an initial population v . This is shown in Fig 4c, which shows the standard deviation of ρ_1^{fit} estimates over 5 replicates with the fitness decay computed from population comprising 10 starting points. Variability between ρ_1^{fit} estimates from the same starting populations are much smaller.

Practically, in order to obtain a global property from empirical landscapes despite their increased heterogeneity, one must sample multiple starting points and consider the ruggedness metric ρ_2 defined via G_μ in Eq (2b). Additionally, the fitted ρ_2^{fit} must be further split into a local estimate ρ_2^{loc} computed from one or a few single-genotype starting populations and a global ρ_2^{glob} computed from many or all possible single-genotype starting populations. Despite the added labour required to measure multiple starting points, such a set up remains scalable as it requires random mutagenesis only, something which is commonly conducted in a pooled format for library generation.

Figure 4d shows how the fitted decay rate ρ_2^{fit} varies as the number of starting points is increased. As it approaches the total number of starting points, the decay rate converges from ρ_2^{loc} (which may vary depending on starting populations) toward ρ_2^{glob} . The difference between ρ_2^{glob} and ρ_2 (dotted) depends on the “spread” of the spectrum; if the power is concentrated in a single frequency then the fitted global value coincides with Eq (1b). Generally, ρ_2^{glob} and ρ_2 are related via a first-order approximation of G_μ (see Methods). Figure 4e shows how ρ_2^{glob} rate varies over the population size, and Fig 4f over the number of generations sampled to obtain the fitness decay. In Fig 4f, the mutation rate is $\theta = 0.1/N$, so that M generations correspond to $\mu = \theta M$ accumulated mutations. Practically, an estimate that approaches ρ_2^{glob} requires sampling 100–1000 starting points with a population size > 500 and at least 15 generations.

Figure 4g compares other ruggedness measures (described in the Supporting Information Section S3) across the empirical landscapes. Given the variation in scaling and linearity between metrics, comparisons should be focused on the ordering of landscapes within each metric, as opposed to comparisons of raw values between metrics. Local epistasis, distance to local maximum, and paths to global maximum agree with the analytical metric ρ_2 in terms of their ordering of all the empirical landscapes tested. Landscape R^2 and roughness to slope ratio agree with the extremities (TEV as the most rugged, ParD3 as the least), however place GB1 and TrpB in a different order. Similarly, ρ_2 groups together TrpB and GB1 landscapes as having a similar, intermediate ruggedness, whereas other methods separate these landscapes, or locate them very close to the TEV landscape. That other ruggedness metrics generally agree with the ordering of ρ_2 on empirical

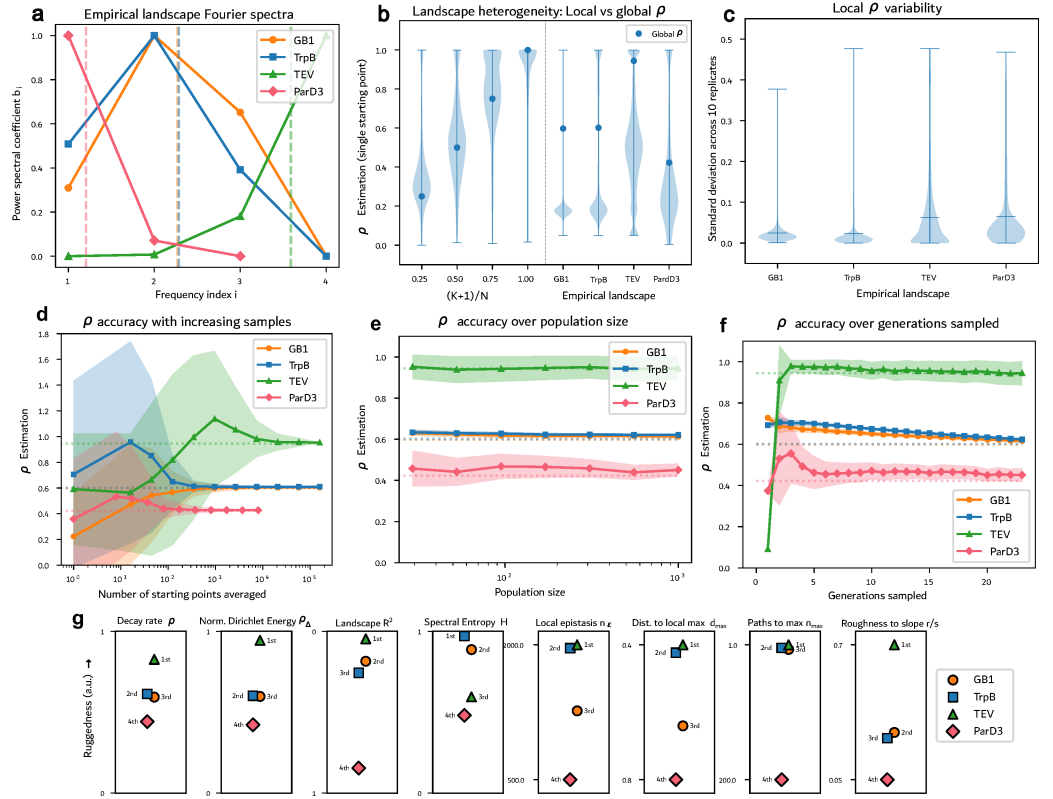


Figure 4. Ruggedness estimation on empirical landscapes. **A:** Fourier spectra of each complete landscape computed using GFT (Methods). Analytical ρ_2 from Eq (1b) indicated by dashed line. **B:** Violin plots showing the distribution of local fitted estimates ρ^{loc} with the fitness decay computed from individual starting points. The dots denote ρ_2 . **C:** Violin plots showing the standard deviation of ρ^{fit} estimates over 5 replicates with the fitness decay computed from 10 starting points. Horizontal bar is the mean standard deviation. Fitted estimates of **D:** ρ_2^{fit} with increasing number of sampled generations, **E:** ρ_2^{glob} with increasing population size and **F:** ρ_2^{glob} with increasing number of sampled generations. Shading is the standard deviation of outcomes across 5 repeats from the same starting points, and if the parameter is not varied, the figures use a population size of 2,500, all starting points, a mutation rate of $\theta = 0.1/N$ and $M = 25$ generations, with the dashed line denoting the analytical ρ_2 value. **G:** Comparison of ruggedness metrics applied to the empirical landscapes.

landscapes provides increased confidence in the metric, which has the major advantage over all others of being experimentally measurable without detailed sequencing data (see Supporting Information Section S5 for a biological interpretation of these ruggedness values).

Optimising DE outcomes with SLIDE

Inferring the ruggedness of a fitness landscape prior to DE allows informed decisions to be made about the DE strategy. In our previous work [10], it was shown that increased population splitting and base chance—a constant probability of selection that is irrespective of fitness—can improve performance on rugged landscapes, but reduces performance on smooth landscapes. By utilising the relationship between ruggedness and fitness decay rate demonstrated in this work, it is therefore possible to optimise DE outcomes without the need for sequencing.

In order to explore the relationship between decay rate and optimal strategy, a large parameter sweep was conducted over 100 different NK landscapes at fixed mutation rate and population

size (Supporting Information Sections S1 and S2). For each landscape, 14,700 DE simulations were conducted over 49 strategy combinations with $N \leq 50$. Fig 5a (dotted line) displays the optimal base chance and splitting values over landscapes of increasing epistatic parameter K for $M = 25$ DE iterations (see Fig S5 for $M > 25$). As $\rho \approx \rho_2 \approx \rho_{NK}$ on NK landscapes, this data can be used as a SLIDE look-up table indexed by ρ_{NK} and queried with a fitted estimate ρ^{fit} or ρ_2^{fit} to select an appropriate combination of DE parameters. To demonstrate SLIDE on NK , for each of the 100 landscapes, 250 decay curves were simulated. From these curves, ρ^{fit} was measured and used to predict optimal parameters, the outcome of which is also shown in Fig 5a (error bars). Note that the optimal strategy is also dependent on the number of generations used in DE—more generations favours more explorative strategies—and the optimal implementation of SLIDE would be to generate this look-up table *in silico* using experimental expectations of generation number and mutation rate (Fig S5).

Two examples of DE curves on NK landscapes are displayed in Fig 5b–c. These figures additionally show how close each selected strategy is to the optimal strategy (yellow), on a heat map of all possible combinations and their performances. Figure 5b displays a smooth landscape ($\rho_{NK} \approx 0.02$), where the strategy selected by SLIDE is highly similar to the baseline strategy (which is already near the optimum). Figure 5c, however, displays a more rugged landscape ($\rho_{NK} \approx 0.55$), on which the baseline strategy is highly sub-optimal and gets trapped at a local optimum within 10 generations. SLIDE, on the other hand, selects the optimal strategy within the strategy space and achieves > 15 % higher fitness than the baseline strategy after 50 generations. Similar data shown in Supporting Information Section S6 demonstrates the robustness of this trend across alternative landscape models to NK (Rough Mount Fuji and a stochastic block model).

A similar approach was then applied to the four combinatorially-complete empirical landscapes GB1, TrpB, TEV and ParD3 (Fig 5d–g) using decay rates estimated from G_{μ} . The strategy look up, as displayed in Fig 5a, was re-generated using NK landscapes of the same dimensions as the landscape: $A = 20$, $N = 4$ (or $N = 3$ for ParD3). As well as this, it was necessary to increase the number of starting points sampled to generate a fitted decay rate estimate that is within 25 % of ρ_2^{glob} . SLIDE out-performed the baseline strategy on all examples, apart from ParD3, for which the baseline strategy was optimal (and therefore SLIDE selected the same parameters). After 150 generations on GB1, TrpB and TEV, it achieved relative fitness gains of 24.9% ($p < 0.001$), 22.6% ($p < 0.01$), and 44.2% ($p < 0.001$) respectively over the baseline approach, which commonly gets trapped at local maxima. These improvements suggest that adapting the DE strategy to the inferred landscape structure can substantially improve optimisation performance. Consequently, integrating the ruggedness estimation framework proposed here with existing DE workflows may yield similar gains compared with fixed-parameter strategies.

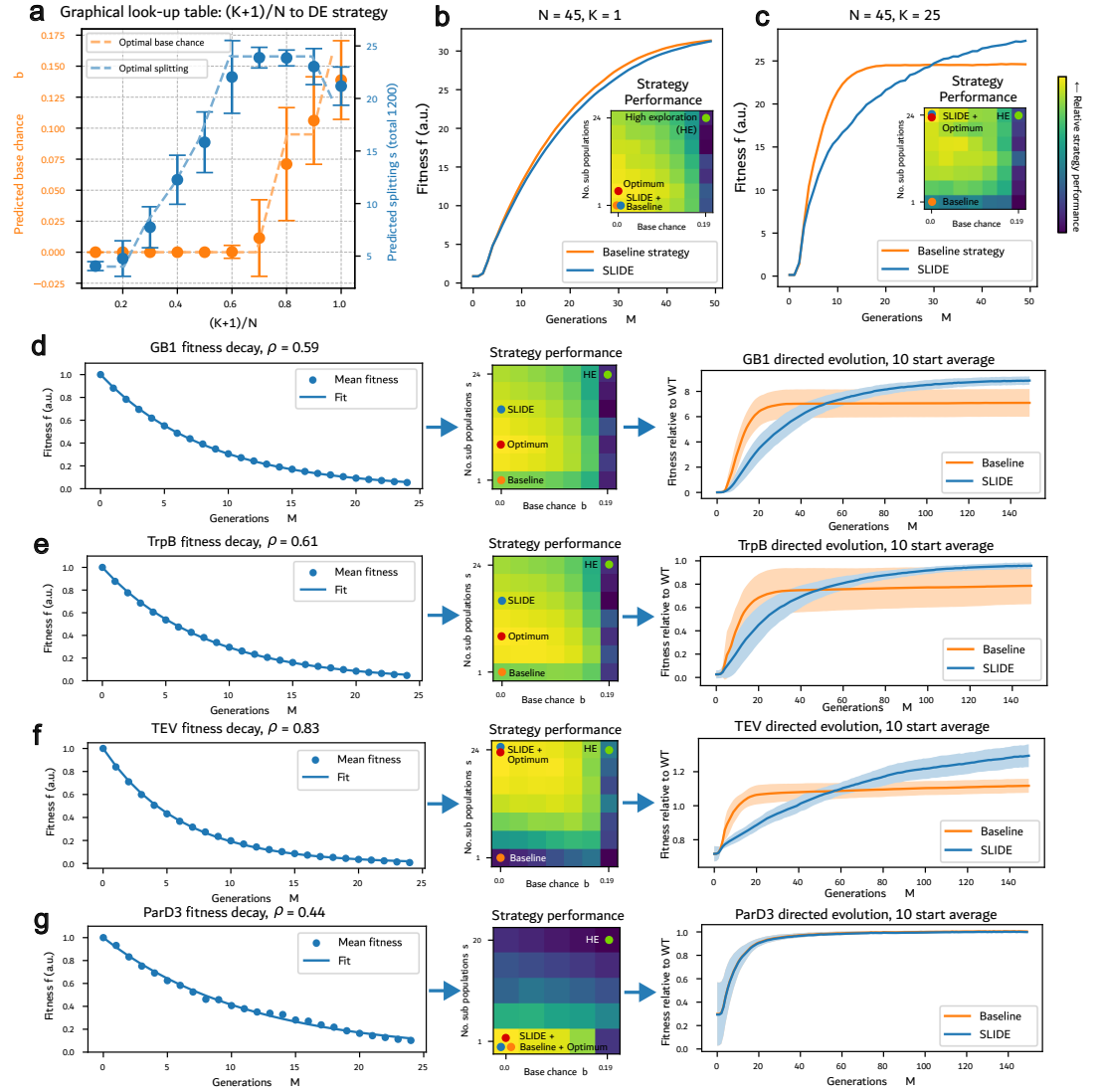


Figure 5. DE using SLIDE on NK and empirical landscapes. **A:** Representation of the SLIDE look up table for NK . Generated from 100 NK landscapes of increasing ruggedness, with 14,700 simulations in total over 49 strategy combinations, $M = 25$ generations and $N \leq 50$. Base chance options = $\{0.00, 0.03, 0.06, 0.10, 0.13, 0.16, 0.19\}$, splitting options = $\{24, 20, 16, 12, 8, 4, 1\}$. Error bars represent the accuracy of SLIDE in selecting the true optimal parameters, from 250 simulations. **B-C:** Implementations of SLIDE on an example smooth and rugged NK landscape, respectively. Heat maps display the strategy selected by each approach, overlaid onto the underlying performance across the whole strategy space (yellow = optimum). Population size = 1200, mutation rate $\theta = 0.1/N$ per generation. Implementations of SLIDE on **D:** GB1, **E:** TrpB, and **F:** TEV. Decay curves from 10,000 starts, mutation rate $\theta = 0.025$ per generation, population size = 2,500. SLIDE strategy look-up table generated for $A = 20, N = 4$. DE simulations conducted from 10 random starts, population size = 1200, mutations = 0.01 per generation. **G:** ParD3. Parameters as above, except base chance options = $[0.00, 0.05, 0.10, 0.14, 0.19]$, splitting options = $[20, 15, 10, 5, 1]$. SLIDE strategy look-up table generated for $A = 20, N = 3$. DE population size = 60. HE = High exploration.

339 Discussion

340 In this paper, we showed that in the absence of selection pressure, the average fitness decay of
341 a population accumulating mutations can be approximated by an exponential curve whose fitted
342 decay rate ρ^{fit} estimates the analytical metric ρ ; the analogous squared-fitness decay rate ρ_2^{fit} esti-
343 mates ρ_2 , which is linked to the normalised Dirichlet energy of the graph Laplacian and quantifies
344 squared fitness variation across mutation neighbours. These decay rates provides practical rugged-
345 ness metrics. Under unbiased mutations, the decay rate ρ_2 reduces to the Fourier power-weighted
346 average frequency of the Hamming graph Laplacian. Under biased mutations, the decay rate is
347 obtained in the eigenbasis of the corresponding mutation operator.

348 Building on these metrics, we introduced SLIDE (Sequence-Free Landscape Inference for Directed
349 Evolution), a framework that estimates landscape ruggedness from average fitness decay curves.
350 Conceptually, SLIDE aligns with broader work employing graph-based methods to extract struc-
351 tural properties of fitness landscapes [3, 25–27]. Unlike existing methods, SLIDE requires only
352 large-scale phenotypic data alongside a mutation rate estimation—removing the need for large-
353 scale sequence-level information. As long as fitness can be coupled to a measurable phenotype
354 such as fluorescence, SLIDE enables ruggedness estimation from bulk population-level measure-
355 ments and knowledge of the average mutation rate. This feature makes SLIDE scalable and exper-
356 imentally accessible.

357 We validated SLIDE *in silico* on synthetic NK landscapes and empirical datasets. Due to homogene-
358 ity, the estimation procedure on NK landscapes is robust to the choice of mutation rate, popula-
359 tion size, decay curve length, and number of starting genotypes, and the analytical, fitted, local,
360 and global decay rates broadly correspond computed from both F_μ and G_μ . The primary require-
361 ment is that the mutation rate is sufficiently large to induce observable decay, yet not so large
362 that observed fitnesses lack any temporal correlation. While the decay rate metric is globally com-
363 patible with structural features such as constant-fitness plateaus, neutral ridges, and zero-fitness
364 genotypes, these features can induce anisotropic local decay behaviour, so that the analytical, fit-
365 ted, local, and global decay rates do not correspond in general. For heterogeneous empirical land-
366 scapes, averaging squared trajectories from approximately 500 to 10,000 starting genotypes (0.25 %
367 to 6.25 % of the landscape, which can be estimated in parallel by approaches described below) was
368 required to obtain ρ_2^{glob} within $\pm 5\%$ of its final value. However, only 3–5 replicates are needed to
369 accurately estimate a local ruggedness metric ρ_2^{loc} . Local metrics have not been further analysed,
370 but could be combined with a downstream DE selection strategy, or used to measure the genetic
371 stability of a genotype.

372 By using SLIDE’s ruggedness estimate to tune DE parameters, we demonstrated up to a 44.2%
373 improvement in DE performance compared to the top-proportion selection baseline, even when

374 ρ_2^{fit} was only within $\pm 25\%$ of ρ_2^{glob} . These gains arise because the inferred ruggedness enables the
375 directed evolution strategy to be adapted to the underlying landscape structure, rather than relying
376 on fixed mutation and selection parameters. While this paper applied DE strategies from [10], the
377 ruggedness metric is broadly compatible with a wide range of advanced DE approaches, including
378 AI-driven workflows. For example, in active learning-assisted DE [14], machine learning techniques
379 are used to guide the selection, balancing exploration of unknown phenotypes with exploitation
380 of high-performing variants. Here, the ruggedness metric obtained from SLIDE could be used to
381 modulate the balance of exploration and exploitation. Similarly, in cluster learning-assisted DE [28],
382 SLIDE could be used to tailor clustering granularity.

383 Although this work focused primarily on the theoretical and computational aspects of SLIDE, the
384 framework is designed to be experimentally implementable. The key requirement is a biological
385 protocol for introducing mutations without applying selection pressure and with a mutation rate
386 that is approximately constant across generations on average. One example is random *in vitro*
387 mutagenesis, which allows targeted exploration of specific regions of the genotype space. Mutant
388 libraries generated in this way can be expressed in cells, with fitness linked to a measurable phe-
389 notype like fluorescence. By iteratively measuring average fluorescence across generations and
390 estimating mutation accumulation (or by simultaneously producing samples with increasing levels
391 of mutation), it becomes possible to construct fitness decay curves and apply SLIDE in a laboratory
392 setting. Such an assay could be conducted across several 96-well plates (or more, if combined with
393 automated pipetting robots), or reduced to a pooled format using mother machine microfluidics.
394 Mother machines are microfluidic chips capable of long-term culture of up to 10^6 individual cells in
395 spatially-segregated chambers that eliminate growth-based competition [29, 30]. In such a format,
396 one need only inject their original variant library and observe cells while inducing mutagenesis, ei-
397 ther with chemical mutagens, UV [31] or even targeted *in vivo* mutagenesis methods [32]. Mutation
398 rate estimation might already be known for well-established workflows, or if not can be estimated
399 by sequencing a sample of variants that is orders of magnitude smaller than the sequencing re-
400 quirements of a combinatorially-complete empirical landscape. Overall, these methods enable
401 practical, low-cost ruggedness estimation across diverse protein families and fitness landscapes.

402 Contrasting the above approaches that focus on fitness measurements that are de-coupled from
403 host/variant growth, the requirement of a selection-free environment could be relaxed by incor-
404 porating selection into the population dynamics. In general, selection can be represented as an
405 additional operator acting on genotype frequencies—e.g. via genotype-dependent growth or ex-
406 plicit enrichment rules—and combined with the mutation operator to yield the mutation-selection
407 dynamics. The observed fitness trajectories are then governed by these combined dynamics, so
408 that SLIDE estimates a ruggedness metric associated with the combined dynamics rather than a
409 freely diffusing population. An approximation is to integrate selection into an effective, biased mu-

410 tation spectrum, such as described in Methods. When selection induces only weak mutation bias,
 411 the resulting metric remains close to the unbiased one. More generally, it measures ruggedness
 412 along the biased dynamics actually used to traverse the landscape. This has two implications for
 413 DE: (i) the same effective spectrum can be used to compute strategy parameters as a function
 414 of the modified ruggedness estimate, and (ii) the decay rate could be identified online during ex-
 415 periments and used as a feedback signal, either to update the ruggedness estimate or to adapt
 416 strategy parameters directly. For instance, under truncation selection, the early fitness gradient
 417 provides a signal that could be linked computationally and analytically to landscape ruggedness
 418 and structure (cf. Fig 5). These extensions would offer new experimental avenues for ruggedness
 419 estimation in settings where mutation and selection act simultaneously, such as continuous *in vivo*
 420 mutagenesis [32] in small-scale bioreactors, and could enable SLIDE to operate *during* DE.

421 In summary, SLIDE offers a novel method for estimating the ruggedness of protein fitness land-
 422 scapes using average phenotypic data alongside mutation rate estimation. These ruggedness esti-
 423 mates can inform experimental strategies in DE, helping to tailor protocols to the underlying struc-
 424 ture of the landscape. Next to DE, SLIDE's ruggedness metric could be extended to characterise
 425 more complex landscapes, such as those underlying synthetic biological circuits. Large-scale map-
 426 ping of ruggedness in such systems could provide a quantitative basis for designing circuits with
 427 greater genetic stability and functional robustness, complementing recent efforts to understand
 428 and engineer the evolutionary potential of synthetic constructs [33].

429 Methods

430 Decay Rate Metric

431 To show why the fitness decay rates ρ and ρ_2 in Eq (2a) can be used as ruggedness metrics, we give
 432 a brief overview of spectral landscape theory. We will represent a population accumulating single-
 433 point mutations as a stochastic signal in a graph, modelling mutagenesis and genotype space. For
 434 a gene of length N , we denote the genotype space by $\mathcal{G} \stackrel{\text{def}}{=} \mathbb{Z}_A^N$ (the space of positive integers up to A ,
 435 in a vector of N entries), containing $|\mathcal{G}| = A^N$ genotypes $\mathbf{x}_1, \dots, \mathbf{x}_{A^N}$. Each genotype can be written
 436 as $\mathbf{x}_i = (\sigma_1^i, \dots, \sigma_N^i)^T$, where $\sigma_j^i \in \mathbb{Z}_A$ is the allele at locus j . For example, if $A = 20$ represents the
 437 number of possible amino acids at each position in a protein of length N , the landscape consists of
 438 A^N distinct genotypes. These genotypes can be represented as nodes in a graph, with each node
 439 being assigned a fitness value $f_i \stackrel{\text{def}}{=} f(\mathbf{x}_i)$, where $f : \mathcal{G} \mapsto \mathbb{R}$ characterises the fitness landscape. A
 440 population is represented by $\mathbf{v} = (v_1, \dots, v_{A^N})^T \in \mathbb{R}^{A^N}$, where each v_i gives the frequency of genotype
 441 $\mathbf{x}_i$. Equivalently, $\mathbf{v}$ can be defined as a probability distribution, in which case $\sum_i v_i = 1$ with all $v_i \in$
 442 $[0, 1]$. The mean population fitness is then computed as $F = \mathbf{f}^T \mathbf{v} = \sum_i f_i v_i$, where $\mathbf{f} = (f_1, \dots, f_{A^N})^T \in$
 443 $\mathbb{R}^{A^N}$ is the stacked vector of fitness values.

444 The edges of the graph connect genotypes that differ by a single mutation, so that every node
 445 is connected to $d \stackrel{\text{def}}{=} N(A - 1)$ other nodes, which is also referred to as a Hamming graph [34].
 446 For node transitions, we define the single-mutation operator $\mathcal{D}$ to map an initial population $\mathbf{v}$ to
 447 $\mathcal{D}(\mathbf{v})$, where every population member has undergone a single, random and equally likely mutation
 448 (see Supporting Information Section S1 for details on our mutation model). It can be shown that
 449 $\mathcal{D}(\mathbf{v}) = D\mathbf{v}$, where $D = D^T \in \mathbb{R}^{A^N \times A^N}$ is a row-stochastic matrix obtained from multiplying the
 450 adjacency matrix of the graph with $1/d$.

451 We assume that the number of mutations that occurs in one mutagenesis round is Poisson dis-
 452 tributed with mean θ , so that the number of mutations X_M occurring between M rounds is Poisson
 453 distributed with mean $M\theta \stackrel{\text{def}}{=} \mu$. The population is therefore expected to evolve as

$$\mathbb{E}_{X_M} [\mathcal{D}^{X_M}(\mathbf{v})] = \left(\sum_{k=0}^{\infty} \Pr(X_M = k) D^k \right) \mathbf{v} = \left(\sum_{k=0}^{\infty} e^{-M\theta} \frac{(M\theta)^k}{k!} D^k \right) \mathbf{v} = \left(\sum_{k=0}^{\infty} e^{-\mu} \frac{\mu^k}{k!} D^k \right) \mathbf{v} \stackrel{\text{def}}{=} \mathcal{M}^M \mathbf{v}. \quad (3)$$

454 Our mutation operator $\mathcal{M}^M$ can be understood as the transition function of a continuous time
 455 Markov chain with a single genotype randomly mutating at rate θ . It can be simplified using matrix
 456 exponentials as

$$\mathcal{M}^M = \sum_{k=0}^{\infty} e^{-\mu} \frac{\mu^k}{k!} D^k = e^{-\mu} \sum_{k=0}^{\infty} \frac{(\mu D)^k}{k!} = e^{-\mu} e^{\mu D} = e^{-\mu(I-D)} \stackrel{\text{def}}{=} e^{-\frac{\mu}{d} \Delta}, \quad (4)$$

457 where we substituted the graph Laplacian $\Delta = \Delta^T \stackrel{\text{def}}{=} d(I - D)$. The mean fitness F_μ of our population
 458 after an average of $\mu = M\theta$ mutations is therefore

$$F_\mu = \mathbf{f}^T (\mathcal{M}^M \mathbf{v}) = \mathbf{f}^T e^{-\frac{\mu}{d} \Delta} \mathbf{v}. \quad (5)$$

459 If necessary, the dependency on the initial population $\mathbf{v}$ can be omitted by taking the expectation
 460 of Eq (5) over many random starting populations. In particular, if $\mathbf{v}$ is sampled uniformly from
 461 single-genotype populations, so that $\mathbb{E}[\mathbf{v}\mathbf{v}^T] = I/A^N$, we can compute $\mathbb{E}_\mathbf{v}[F_\mu^2]$ as:

$$G_\mu \stackrel{\text{def}}{=} \mathbb{E}_\mathbf{v}[F_\mu^2] = \mathbf{f}^T e^{-\frac{\mu}{d} \Delta} \mathbb{E}_\mathbf{v}[\mathbf{v}\mathbf{v}^T] e^{-\frac{\mu}{d} \Delta^T} \mathbf{f} = \frac{1}{A^N} \mathbf{f}^T e^{-\frac{2\mu}{d} \Delta} \mathbf{f}. \quad (6)$$

462 If $\mathbf{v}$ is sampled from a smaller subset with $\mathbb{E}[\mathbf{v}\mathbf{v}^T] = \Sigma_\mathbf{v}$, $G_\mu = \mathbf{f}^T e^{-\frac{\mu}{d} \Delta} \Sigma_\mathbf{v} e^{-\frac{\mu}{d} \Delta^T} \mathbf{f}$. Eqs (5) and (6) can be
 463 further simplified by considering the special structure of a Hamming graph. Its real-valued, sym-
 464 metric Laplacian admits a **unitary Fourier basis as eigenbasis** [35], so that Δ can be factorised as $\Delta =$
 465 $P\Lambda P^*$, where $P \in \mathbb{C}^{A^N \times A^N}$ satisfying $P^* P = I$ contains the eigenvectors, and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_{A^N}) \in$
 466 $\mathbb{R}^{A^N \times A^N}$, with the eigenvalues ordered as $\lambda_1 \leq \lambda_2 \leq \dots \lambda_{A^N}$. Analogous to the standard Fourier
 467 transform, the eigenvalues capture a frequency and the columns of P eigenfunctions, each of
 468 which can be interpreted as eigenlandscapes of increasing ruggedness for increasing λ_i . For the
 469 Hamming graph Laplacian, there are $N + 1$ unique eigenvalues with value $\lambda_i = A \times i$ and multi-

470 plicity $\binom{N}{i}(A-1)^i$, $i = 0, 1, \dots, N$ [36]. The matrix P may be chosen as $P_{j,i} \stackrel{\text{def}}{=} \chi_j(x_i)/\sqrt{A^N}$, where
 471 $\chi_j(x_i) \stackrel{\text{def}}{=} \exp(2\pi i \langle r_j, x_i \rangle / A)$ is the Fourier basis function, $r_j \in \mathcal{G}$ the frequency index, and $\langle r_j, x_i \rangle$ the
 472 dot product defined on $\mathcal{G}$ (dot product modulo A) [35]. For $\lambda_0 = 0$, the eigenlandscape is a con-
 473 stant equal to the landscape average $\bar{f} \stackrel{\text{def}}{=} \mathbf{1}^\top \mathbf{f} / A^N$, where $\mathbf{1} \stackrel{\text{def}}{=} (1, \dots, 1)^\top$. For $\lambda_1 = A$, the land-
 474 scape is linear, i.e. depends only on the allele σ_j at a single locus j (Fig 2c). For larger eigenvalues,
 475 the eigenlandscapes are rugged, i.e. they depend on the allele at multiple loci (exactly i differ-
 476 ent loci for $\lambda_i = Ai$). From these eigenvalue-eigenvector pairs, it also follows that $\lim_{\mu \rightarrow \infty} F_\mu = \bar{f}$
 477 and $\lim_{\mu \rightarrow \infty} G_\mu = \mathbf{f}^\top \mathbf{1} \mathbf{1}^\top \mathbf{f} / (A^N)^2 = \bar{f}^2$. In other words, if we sample uniformly from all single-
 478 genotype populations and average the fitness trajectories via $\mathbb{E}_v[F_\mu^2]$, the resulting G_μ decays from
 479 $G_0 = \mathbf{f}^\top \mathbf{f} / A^N$ exponentially to $\bar{f}^2$ as $\mu \rightarrow \infty$.

480 The special eigenbasis of the Hamming graph allows the mean fitness decay to be broken down
 481 into contributions of different frequencies. Substituting $\Delta = P \Lambda P^*$ in Eqs (5) and (6) and defining
 482 $\hat{\mathbf{f}} \stackrel{\text{def}}{=} P^* \mathbf{f}$ and $\hat{\mathbf{v}} \stackrel{\text{def}}{=} P^* \mathbf{v}$ yields:

$$F_\mu = \hat{\mathbf{f}}^* e^{-\frac{\mu}{d} \Lambda} \hat{\mathbf{v}} = \sum_{i=0}^N e^{-\frac{\mu \lambda_i}{d}} \left(\sum_{j \in \mathcal{I}_i} \hat{\mathbf{f}}_j^* \hat{\mathbf{v}}_j \right) \stackrel{\text{def}}{=} \sum_{i=0}^N e^{-\frac{\mu \lambda_i}{d}} a_i, \quad (7a)$$

$$G_\mu = \frac{1}{A^N} \hat{\mathbf{f}}^* e^{-\frac{2\mu}{d} \Lambda} \hat{\mathbf{f}} = \frac{1}{A^N} \sum_{i=0}^N e^{-\frac{2\mu \lambda_i}{d}} \left(\sum_{j \in \mathcal{I}_i} |\hat{\mathbf{f}}_j|^2 \right) \stackrel{\text{def}}{=} \frac{1}{A^N} \sum_{i=0}^N e^{-\frac{2\mu \lambda_i}{d}} b_i, \quad (7b)$$

483 where $\mathcal{I}_i$ ($|\mathcal{I}_i| = \binom{N}{i}(A-1)^i$) groups indices associated with the i th eigenvalue and

$$a_i \stackrel{\text{def}}{=} \sum_{j \in \mathcal{I}_i} \hat{\mathbf{f}}_j^* \hat{\mathbf{v}}_j, \quad b_i \stackrel{\text{def}}{=} \sum_{j \in \mathcal{I}_i} |\hat{\mathbf{f}}_j|^2. \quad (8)$$

484 Because $\lambda_0 = 0$ and $\lambda_i > 0$ for $i \geq 1$, it holds that

$$\lim_{\mu \rightarrow \infty} F_\mu = a_0 = \frac{(\mathbf{1}^\top \mathbf{v})(\mathbf{1}^\top \mathbf{f})}{A^N} = \bar{f}, \quad \lim_{\mu \rightarrow \infty} G_\mu = \frac{b_0}{A^N} = \frac{(\mathbf{1}^\top \mathbf{f})^2}{(A^N)^2} = \bar{f}^2. \quad (9)$$

485 Thus $b_0 = A^N \bar{f}^2$ under the Fourier normalisation used here. Expressions written in Fourier-power
 486 units subtract b_0 , while equivalent expressions written in genotype-vector units subtract $A^N \bar{f}^2$. The
 487 operation $P^* \mathbf{f}$ can be understood as a coordinate transformation and the vector $\hat{\mathbf{f}}$ as the landscape
 488 $\mathbf{f}$ represented in the Fourier domain. The coefficients $b_0, \dots, b_N$ are obtained from squaring the
 489 modulo of Fourier coefficients and therefore represent the power or energy per frequency λ_i . The
 490 power spectra of several NK landscapes are shown in Fig 7a. For the less rugged landscapes, the
 491 power spectrum peaks at a lower frequency than for the more rugged landscapes.

492 Equations (7a) and (7b) show that as the accumulated mutations μ increase, the contribution of
 493 each Fourier component to the measured fitness decreases at an exponential rate proportional to
 494 λ_i , such as shown in Fig 7b. For example, for a linear, zero-mean landscape for which only $b_1 \neq 0$,
 495 the mean fitness F_μ will decay at a rate proportional to $\lambda_1 = A$ per accumulated mutation, whereas

for a maximally rugged landscape for which only $b_N \neq 0$, it will decay at a rate proportional to $\lambda_N = AN$, which is N times faster than the linear one. For most landscapes, the decay will be composed from multiple exponentials, such as for the NK landscapes in Figs 7b–c.

Although the coefficients $a_0, \dots, a_N$ and $b_0, \dots, b_N$ could be estimated from Eqs (7a) and (7b), their estimates can become inaccurate (see following section) if F_μ and G_μ are noisy. Assuming that the coefficients a_i and b_i from Eqs (7a) and (7b) vary significantly in magnitude, the fitness decay curve is dominated by the larger-magnitude exponentials. In case of a single dominant component, Eqs (5) and (6) can be approximated by $F_\mu \approx a_0 + e^{-\frac{\mu\lambda_{i^*}}{d}} a_{i^*}$ and $G_\mu \approx (b_0 + e^{-\frac{2\mu\lambda_{i^*}}{d}} b_{i^*})/A^N$, where i^* is the index of the dominant component. We therefore seek to approximate Eq (7a) and Eq (7b) using a single exponential as

$$F_\mu \approx Ce^{-\mu\rho} + c, \quad (10a)$$

$$G_\mu \approx C_2e^{-2\mu\rho_2} + c_2, \quad (10b)$$

where from Eq (9), it holds that $c = \bar{f}$ and $c_2 = \bar{f}^2$. We introduce the decay rates ρ and ρ_2 as scalar ruggedness metrics that approximate the average of the frequencies λ_i weighted by a_i or b_i . This can be shown by expanding the right-hand sides of Eqs (10a) and (10b) as

$$Ce^{-\mu\rho} + c = C(1 - \mu\rho + \frac{1}{2}\mu^2\rho^2 + \dots) + c = C + c - C\mu\rho + \mathcal{O}(\mu^2), \quad (11a)$$

$$C_2e^{-2\mu\rho_2} + c_2 = C_2(1 - 2\mu\rho_2 + 2\mu^2\rho_2^2 + \dots) + c_2 = C_2 + c_2 - 2C_2\mu\rho_2 + \mathcal{O}(\mu^2), \quad (11b)$$

and the right-hand sides of Eqs (7a) and (7b) as

$$\sum_{i=0}^N e^{-\frac{\mu\lambda_i}{d}} a_i = \sum_{i=0}^N a_i - \frac{\mu}{d} \sum_{i=0}^N a_i \lambda_i + \mathcal{O}(\mu^2), \quad (12a)$$

$$\frac{1}{A^N} \sum_{i=0}^N e^{-\frac{2\mu\lambda_i}{d}} b_i = \frac{1}{A^N} \sum_{i=0}^N b_i - \frac{1}{A^N} \frac{2\mu}{d} \sum_{i=0}^N b_i \lambda_i + \mathcal{O}(\mu^2) \quad (12b)$$

Ignoring higher-order terms, comparing coefficients yields $C + c = \sum_{i=0}^N a_i$ and $C\mu\rho = \mu/d \sum_{i=0}^N a_i \lambda_i$ for F_μ , and $C_2 + c_2 = \sum_{i=0}^N b_i/A^N$ and $C_2\mu\rho_2 = \mu/d \sum_{i=0}^N b_i \lambda_i/A^N$ for G_μ . We also note that since $\lambda_0 = 0$, Eq (7) and Eqs (9)–(10) imply that $c = a_0 = \bar{f}$ for F_μ and $c_2 = b_0/A^N = \bar{f}^2$ for G_μ , so that

$$\rho \approx \frac{1}{d} \frac{\sum_{i=0}^N a_i \lambda_i}{\sum_{i=0}^N a_i - a_0} = \frac{1}{d} \frac{\hat{\mathbf{f}}^* \Lambda \hat{\mathbf{v}}}{\hat{\mathbf{f}}^* \hat{\mathbf{v}} - a_0} = \frac{1}{d} \frac{\mathbf{f}^T \Delta \mathbf{v}}{\mathbf{f}^T \mathbf{v} - \bar{f}}, \quad (13a)$$

$$\rho_2 \approx \frac{1}{d} \frac{\sum_{i=0}^N b_i \lambda_i}{\sum_{i=0}^N b_i - b_0} = \frac{1}{d} \frac{\hat{\mathbf{f}}^* \Lambda \hat{\mathbf{f}}}{\hat{\mathbf{f}}^* \hat{\mathbf{f}} - b_0} = \frac{1}{d} \frac{\mathbf{f}^T \Delta \mathbf{f}}{\mathbf{f}^T \mathbf{f} - A^N \bar{f}^2}. \quad (13b)$$

The right-hand side of Eq (13b) highlights that the analytical decay rate ρ_2 for G_μ is the Dirichlet energy $\mathbf{f}^T \Delta \mathbf{f}$ normalised by $\mathbf{f}^T \mathbf{f} - A^N \bar{f}^2$, equivalently $\mathbf{f}^T \mathbf{f} - b_0$. Equation (13b) also shows that ρ_2 is related to other spectral measures such as the R^2 measure, $R^2 \triangleq \sum_{i=2}^N b_i / (\sum_{j=1}^N b_j)$, or the spectral

516 entropy, $H \stackrel{\text{def}}{=} -\sum_{i=1}^N p_i \log(p_i)$, where $p_i \stackrel{\text{def}}{=} b_i / (\sum_{j=1}^N b_j)$. However, note that Eq (13b) explicitly takes
 517 the mutation spectrum into account, whereas other measures do not.

518 From the first-order approximation in Eq (11), the analytical metric ρ_2 characterises the G_μ decay
 519 for small μ . Additionally, Jensen's inequality for convex functions can be used to show that

$$G_\mu = \frac{b_0}{A^N} + \frac{B}{A^N} \sum_{i=1}^N \frac{b_i}{B} e^{-\frac{2\mu\lambda_i}{d}} \geq \frac{b_0}{A^N} + \frac{B}{A^N} e^{-2\mu\rho_2}, \quad (14)$$

520 where $B = \sum_{i=1}^N b_i$, i.e. the single exponential decaying at rate ρ_2 provides a lower bound.

521 We note that ρ for F_μ depends on the initial population v , whereas for a sufficiently large number
 522 of samples in Eq (6), ρ_2 for G_μ reflects a global property of the landscape if computed from all
 523 starting points. We therefore distinguish between analytical decay rates (ρ , ρ_2), fitted decay rates
 524 (ρ^{fit} , ρ_2^{fit}), and their local (ρ^{loc} , ρ_2^{loc}) and global (ρ^{glob} , ρ_2^{glob}) variants. A summary of the notation used
 525 is provided in Table 2, including the decay rates for biased mutation spectra introduced in the
 526 following section. If a single component a_{i^*} or b_{i^*} is dominating, then the corresponding decay
 527 rates satisfy approximately

$$\rho \approx \rho_2 \approx \frac{\lambda_{i^*}}{d} \leq \frac{A}{A-1}, \quad (15)$$

528 with fitted decay rates being approximately equal to the one in Eq (15).

529 The approximation from Eqs (10a) and (10b) can be analysed in more detail for house-of-cards and
 530 NK landscapes (see Supporting Information S4), where it can be proved that the approximations
 531 hold increasingly well in the limit $N \rightarrow \infty$ [37].

Table 2. Summary of ruggedness and decay-rate notation. Column F_μ denotes metrics computed from the average population fitness decay, whereas column G_μ denotes metrics computed from the squared average population fitness decay.

F_μ	G_μ	Description
ρ	ρ_2	Analytical decay rate.
ρ^{fit}	ρ_2^{fit}	Fitted decay rate.
ρ^{loc}	ρ_2^{loc}	Local fitted decay rate.
ρ^{glob}	ρ_2^{glob}	Global fitted decay rate.
ρ_{NK}		Closed-form NK ruggedness value.
ρ_{HoC}		Closed-form HoC ruggedness value.
	$\tilde{\rho}_2$	Decay rate for weighted, undirected mutation operator.
	$\bar{\rho}_2$	Decay rate for directed mutation operator.

532 Biased Mutation Spectra

533 The single-mutation operator D allows the fitness decay to be related to the Hamming graph Lapla-
 534 cian, which is amenable to a Fourier decomposition. In this scenario, all mutations are equally
 535 likely, and each population member evolves according to an unbiased random walk on a regular,
 536 undirected graph (Fig 6a). However, in real-world scenarios, the mutation spectra are biased, e.g.

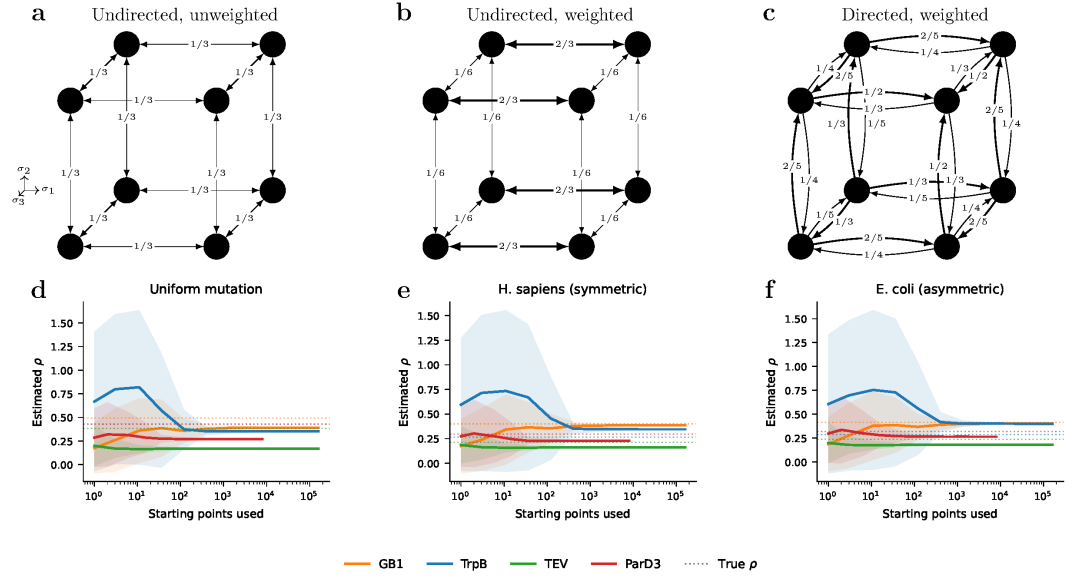


Figure 6. Impact of mutation model on ruggedness inference. **a-c:** Graph representations of mutation models ($A = 2$, $N = 3$) with edge values denoting transition probabilities. **d-f:** Ruggedness estimation against number of starting points used on empirical landscapes. The mutation models correspond to cases **a-c**. Shaded regions denote the standard deviation of the estimate across different random starting points. The estimates are computed as in Fig 4a, with mutation rate $\theta = 0.1/N$ per generation, population size = 2,500, and $M = 75$ generations. The mutation models for **b** and **c** were extracted from [38] and the simulation was extended with mapping from codon to amino acids. For details on the mutation models used, see Supporting Information Section S7.

537 due to transition/transversion preferences, context effects, or codon usage. Also natural selection,
 538 e.g. due to overexpressed proteins, can be modelled as mutation bias. These effects can result
 539 in biased walks on weighted, directed or undirected graphs (Figs 6b-c), whose Laplacians are *not*
 540 amenable to a Fourier decomposition. However, the fitness decay can still be used to infer a land-
 541 scape metric associated with the corresponding mutation model.

542 Suppose that the graph is weighted but undirected with mutation operator $\tilde{D}$ and symmetric Lapla-
 543 cian $\tilde{\Delta}$, which occurs when e.g. all transition mutations are preferred over transversion mutation
 544 (Fig 6b). Since $\tilde{\Delta} = \tilde{\Delta}^T$, the equations of the previous section remain valid. In particular, the matrix
 545 identity from Eq (6), $\exp(-\frac{\mu}{d}\tilde{\Delta})\exp(-\frac{\mu}{d}\tilde{\Delta}^T) \equiv \exp(-\frac{2\mu}{d}\tilde{\Delta})$ holds, which requires $\tilde{\Delta}$ and $\tilde{\Delta}^T$ to commute.
 546 However, the analytical squared-decay metric $\tilde{\rho}_2$ approximates the normalised Dirichlet energy on
 547 the weighted graph instead:

$$\tilde{\rho}_2 \approx \frac{1}{d} \frac{\sum_{i=1}^N \tilde{b}_i \tilde{\lambda}_i}{\sum_{i=1}^N \tilde{b}_i} = \frac{1}{d} \frac{\mathbf{f}^T \tilde{\Delta} \mathbf{f}}{\mathbf{f}^T \mathbf{f} - \tilde{b}_0}, \quad (16)$$

548 where $\tilde{b}_i$ and $\tilde{\lambda}_i$ are obtained from the eigendecomposition of the weighted Laplacian, $\tilde{\Delta} = \tilde{P} \tilde{\Lambda} \tilde{P}^*$.
 549 The metrics ρ_2 and $\tilde{\rho}_2$ are associated with particular mutation operators and may differ in general.
 550 A landscape may appear rugged via ρ_2 and smooth via $\tilde{\rho}_2$. Letting $E_{\tilde{\Delta}} \triangleq \Delta - \tilde{\Delta} = d(\tilde{D} - D)$, the

551 difference $|\rho_2 - \bar{\rho}_2|$ can be bounded as

$$|\rho_2 - \bar{\rho}_2| = \frac{1}{d} \frac{|\mathbf{f}^\top E_{\bar{\Delta}} \mathbf{f}|}{|\mathbf{f}^\top \mathbf{f} - b_0|} \leq \frac{1}{d} \frac{\|\mathbf{f}\|_2^2 \|E_{\bar{\Delta}}\|_2}{|\mathbf{f}^\top \mathbf{f} - b_0|}, \quad (17)$$

552 where $\tilde{b}_0 = b_0 = A^N \bar{f}^2$ assuming that for both weighted and directed settings, the mutation kernels
 553 are doubly stochastic and irreducible, so that the fitness decay always converges to the (squared)
 554 landscape average. Thus the metrics are close when the mutation kernel perturbation $\|D - \bar{D}\|_2$ is
 555 small.

556 For a directed graph, let the non-symmetric mutation operator and Laplacian be $\bar{D}$ and $\bar{\Delta}$, respec-
 557 tively. This represents the most general setting, which can occur e.g. under selection pressure
 558 or using *in vivo* mutagenesis methods [39]. In this setting, the population still experiences an ex-
 559 ponential fitness decay and the derivations of the previous section remain valid until Eq (6), but
 560 without commutativity of $\bar{\Delta}$ and $\bar{\Delta}^\top$, the expression $\exp(-\frac{\mu}{d} \bar{\Delta}) \exp(-\frac{\mu}{d} \bar{\Delta}^\top)$ cannot be simplified. In-
 561 stead, consider the first-order approximation $\exp(-\frac{\mu}{d} \bar{\Delta}) \approx I - \frac{\mu}{d} \bar{\Delta}$, which substituted in Eq (6) yields

$$\bar{G}_\mu \approx \frac{1}{A^N} \mathbf{f}^\top \left(I - \frac{2\mu}{d} \frac{\bar{\Delta} + \bar{\Delta}^\top}{2} \right) \mathbf{f} + \mathcal{O}(\mu^2), \quad (18)$$

562 where $\bar{\Delta}_{\text{sym}} \stackrel{\text{def}}{=} (\bar{\Delta} + \bar{\Delta}^\top)/2$ is the orthogonal projection (via Frobenius norm) of $\bar{\Delta}$ onto the subspace
 563 of symmetric matrices. Comparing coefficients with Eq (11) results in the approximation

$$\bar{\rho}_2 \approx \frac{1}{d} \frac{\mathbf{f}^\top \bar{\Delta}_{\text{sym}} \mathbf{f}}{\mathbf{f}^\top \mathbf{f} - \bar{b}_0}, \quad (19)$$

564 where it can be assumed that $\bar{b}_0$ again equals A^N times the squared landscape average when the
 565 mutation kernel has the uniform stationary distribution. As for the weighted, undirected setting,
 566 the difference $|\rho_2 - \bar{\rho}_2|$ may be upper-bounded using the perturbation $E_{\bar{\Delta}} \stackrel{\text{def}}{=} \Delta - \bar{\Delta}_{\text{sym}}$ such as in
 567 Eq (17).

568 To analyse the impact of biased mutations qualitatively, the ruggedness estimation from Fig 4a is
 569 repeated for different mutation models in Figs 6d–f on the empirical landscapes GB1, TrpB, TEV,
 570 and ParD3. The figures show the fitted estimates ρ_2^{fit} over the number of starting points (solid lines),
 571 with the shaded area corresponding to the standard deviation and the dashed lines denoting the
 572 analytical squared-decay metrics ρ_2 (d) and $\bar{\rho}_2$ (e–f). Compared to Fig 4a, an additional DNA to amino
 573 acid model is included for Figs 6d–f. As a consequence, the landscape spectra are more spread out
 574 and the difference between the lower bound of the exponential and the fitted exponential is larger.
 575 The mutation biases are taken from [38] (see Supporting Information Section S7).

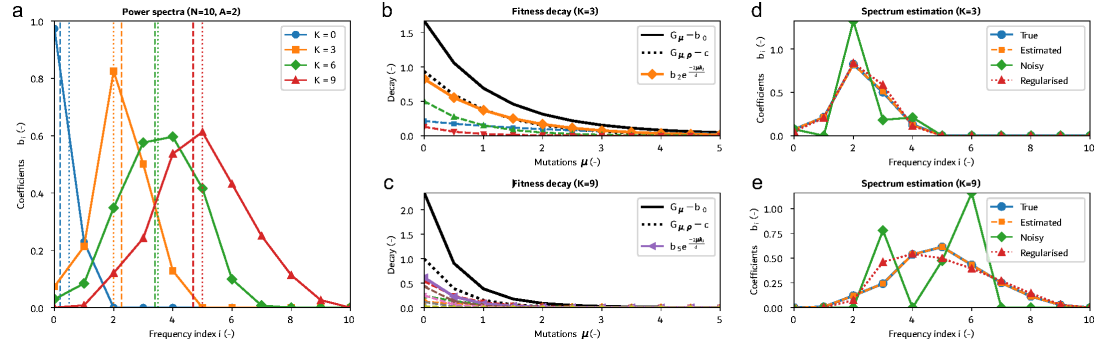


Figure 7. Power spectrum estimation on NK landscapes ($N = 10, A = 20$). **A:** Power spectra of NK landscapes for $K \in \{0, 3, 6, 9\}$. The plot also shows the maximum power frequency from Eq (S2) (dotted) and the frequency obtained from decay-rate estimation (dashed). **B–C:** Decay rates sampled from the landscapes from Fig 7a for $K = 3$ (top) and $K = 9$ (bottom). The figure shows the decay as $G_\mu - b_0/A^N$, the fitted curve $G_{\mu, \rho_2^{\text{fit}}} - c_2$ from Eq (24), and the components $b_i \exp(-2\mu\lambda_i/d)/A^N$ of the five largest b_i of the decay decomposition, with the legend highlighting the dominant one (cf. Fig 7a). **D–E:** Estimation of the full power spectrum via least squares (Eq (20)) and regularised least squares (Eq (22)), with and without noise. The regularisation reduces the impact of noise.

576 Estimating power spectra and ruggedness parameters

577 Generally, the power spectrum of a fitness landscape is composed from both low-frequency and
 578 high-frequency terms, such as shown in Fig 7a. Practically, if the landscape f is known, its power
 579 spectrum can be computed by reshaping f as an N -dimensional cube of size A along each axis,
 580 applying an N -dimensional Fast Fourier Transform, and grouping the terms as in Eq (7b).

581 If the landscape is not known, the power spectral coefficients b_i from Eq (8) can be estimated
 582 using the decay curve G_μ from Eq (7b). The procedure for estimating a_i from F_μ is similar. Let
 583 $G = (G_0, G_{\mu_1}, \dots, G_{\mu_M})^T$ be the squared average fitness measurements of populations accumulating
 584 $0, \mu_1, \dots, \mu_M$ mutations and define $b = (b_0 \dots, b_N)^T$. The following least squares problem can be used
 585 to estimate b from fitness measurements G :

$$b = \arg \min_{x \in \mathbb{R}_{\geq 0}^{N+1}} \|G - \Phi x\|_2^2, \quad (20)$$

586 where element i, j of the matrix $\Phi \in \mathbb{R}^{(M+1) \times (N+1)}$ is defined as

$$\Phi_{i,j} = e^{-\frac{2\mu_i \lambda_j}{d}}, \quad (21)$$

587 for $i = 0, \dots, M$ and $j = 0, \dots, N$. For noise-free observations, problem (20) can be solved using stan-
 588 dard techniques to accurately determine the fitness landscape spectrum b (Fig 7d–e). However, ad-
 589 dition of noise with a signal-to-noise ratio ≤ 10 can already significantly perturb the estimate from
 590 Eq (20), which is caused by the ill-conditioned Φ that amplifies the estimation error. To improve

the robustness of the estimate, Eq (20) can be extended with a regularisation term as

$$b = \arg \min_{x \in \mathbb{R}_{\geq 0}^{N+1}} \|G - \Phi x\|_2^2 + \gamma \|x\|_2^2, \quad (22)$$

where $\gamma \geq 0$ is a regularisation parameter. While the results from Fig 7d–e show that regularisation can improve the estimation accuracy, it depends on having chosen γ appropriately; choosing a poor γ will lower accuracy. The inference method could be developed further, incorporating more knowledge about the distribution of b and the noise process, which is beyond the scope of this paper.

Instead of estimating the full spectrum, we estimate fitted ruggedness parameters using the approximations from Eq (10). The ρ_2^{fit} obtained from fitting G_μ will in general not equal the analytical ρ_2 from Eq (13b); according to Eq (14), the analytical ρ_2 defines a lower-bound single exponential. If the power spectrum is concentrated in a single frequency (such as for NK landscapes), then $\rho_2^{\text{fit}} \approx \rho_2$. The scalar parameters C_2 , ρ_2^{fit} , and c_2 are unknown *a priori* and must be estimated from data. As for the power spectrum estimation, the squared average fitness values G are recorded for a population accumulating $\mu_1, \dots, \mu_M$ mutations. We then seek to find C_2 , ρ_2^{fit} , and c_2 such that

$$G_{\mu_i} \approx C_2 e^{-2\mu_i \rho_2^{\text{fit}}} + c_2. \quad (23)$$

As for the full spectrum estimation, the mutations are stochastic and the decay curves are noisy, which impacts the parameter estimation process. To simplify the estimation, we assume that G_0 is noise free, and that the curve fitted via Eq (23) must pass through it, which allows us to discard parameter C_2 by setting $C_2 = G_0 - c_2$. For the remaining data points, we seek to find ρ_2^{fit} and c_2 that minimize the sum of squares error:

$$\min_{\rho_2^{\text{fit}}, c_2 \in \mathbb{R}_{\geq 0}} \sum_{i=1}^M \left((G_0 - c_2) e^{-2\mu_i \rho_2^{\text{fit}}} - (G_{\mu_i} - c_2) \right)^2. \quad (24)$$

The procedure for estimating ρ^{fit} from F_μ is similar and requires discarding the factor 2 from exponentials. As long as $c_2 \neq 0$, Eq (24) is a nonlinear optimisation problem that cannot be solved using standard least squares techniques and may have multiple local optima. Here, we are using standard scientific packages for constrained non-linear optimisation (specifically, the `curve_fit` function from the `scipy.optimize` package [40]), which returns a solution within instants (≤ 1 s) on a standard laptop. Initialising the routine with $(\rho_2^{\text{fit}})_{\text{init}} = 0.5$ and $(c_2)_{\text{init}} = G_M$ and constraining the optimisation variables by $\rho_2^{\text{fit}} \in [0, A/(A-1)]$ and $c_2 \in [0.5 \times (c_2)_{\text{init}}, 1.5 \times (c_2)_{\text{init}}]$ (see Code Availability), this procedure is robust to noise and could be extended with uncertainty weights (regularisation). The curve fitting for two NK landscapes is shown in Fig 7b–c, with the fitted decay rates ρ_2^{fit} shown in Fig 7a. For these examples, the power spectrum peaks at a particular frequency, so that Eq (23) results in a good approximation of the true decay composed from multiple exponentials.

620 Author Contributions

621 **Sebastian Towers** Responsible for conceptualisation, experimental design, code for simulations,
622 production of figures and writing.

623 **Jessica James** Responsible for conceptualisation, experimental design, code for simulations, pro-
624 duction of figures and writing.

625 **Harrison Steel** Contributions to conceptualisation and editing.

626 **Idris Kempf** Project management, contributions to conceptualisation, experimental design, code
627 for simulations, writing and editing.

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724 Supporting Information

725 **S1 Text. Supplementary Information.** Contains Sections S1–S7 describing simulation methods,
726 ruggedness metrics, additional results, and robustness analyses for both synthetic and empirical
727 fitness landscapes.

728 **S2 Fig. Comparison of strategy space and decay curve trends across different landscape**
729 **models.** Heat maps show strategy spaces across low, mid and high fitness implementations of
730 the *NK*, Rough Mount Fuji and stochastic block models. White line tracks the optimal strategy as
731 ruggedness increases. Each heat map represents an average across 300 starting points. Line plots
732 on the right hand side display the average fitness decay over the same landscapes represented in
733 the heat maps.

734 **S3 Fig. Replication of Fig 3 using biased mutation spectra on *NK* landscapes.** The ρ^{fit} values
735 are estimated across landscapes of increasing ruggedness using 25 generations and a mutation
736 rate $\theta = 0.5$ per generation. The figure shows mean and standard deviation for 300 estimates.

737 **S4 Fig. Decay curves on empirical landscapes.** The figures show decay curves sampled from
738 empirical landscapes, fitted decay curves, and the idealised decay curves, for unbiased mutations
739 (top), biased symmetric mutations (middle), and biased asymmetric mutations (bottom). To imple-
740 ment biased mutations, all landscapes incorporate a DNA-to-codon mapping. True c represents
741 the true fitness stationary distribution. For the symmetric and uniform mutation models, this is the
742 unweighted landscape average. For the asymmetric mutation model, this is calculated based on
743 the stationary state of the codon transition matrix. Fitted c is the constant term in the exponential
744 fit to the observed data. The mutation rate per generation is $\theta = 0.1/N$.

745 **S5 Fig. Look up table for *NK* with higher numbers of DE iterations.** Parameters as in Fig 3a
746 but with $M = \mathbf{A}: 25, \mathbf{B}: 50$ and $\mathbf{C}: 100$ generations. Error bars represent the accuracy of SLIDE in
747 selecting the true optimal parameters, from 250 simulations.

750 **S1 Simulating mutagenesis and selection**

751 In our simulations, we represent a population of P cells at iteration k as $\mathbf{p}_k = (\mathbf{x}_k^1, \dots, \mathbf{x}_k^P)$, where
752 $\mathbf{x}_k^i \in \mathcal{G}$ is the genotype of cell i . Given an empirical or theoretical landscape f , each genotype is
753 associated with a fitness $f_k^i = f(\mathbf{x}_k^i)$. When the population $\mathbf{p}_k$ is mutagenised, each member $\mathbf{x}_k^i$
754 is exposed to X single mutations to yield $\mathbf{x}_{k+1}^i$, where X is an independent identically distributed
755 Poisson random variable. To obtain the decay curves used for ruggedness parameter estimation,
756 we repeat the mutagenesis process for M iterations and record the average population fitness
757 $F_k = 1/P \sum_{i=1}^P f_k^i$, its square $G_k = F_k^2$, and the average accumulated mutations μ_k . This process is
758 repeated and averaged over different starting populations $\mathbf{p}_0$ to generate a decay curve such as
759 the one shown in Fig 3A.

760 For the selection process used in directed evolution (DE) experiments, each iteration is extended
761 with a selection procedure. Here, we use an approach developed in [10] that parameterises the se-
762 lection using probabilistic “selection functions”. The population members $\mathbf{x}_k^i$ are ranked according
763 to their fitness f_k^i . A fraction of cells is then randomly selected from $\mathbf{p}_k$ according to the probability
764 distribution shown in Fig 1 (top right). This distribution is parameterised by two parameters: the
765 base chance b and threshold t . If a cell’s fitness f_k^i is in the top t percentile (within the population
766 $\mathbf{p}_k$), it is guaranteed to be selected for iteration $k+1$. Otherwise, the selection probability is b , which
767 is independent between cells. Here, the fraction of cells selected at each iteration is kept constant
768 as $f = 0.2$, fixing t as $t = (1 - f)/(1 - b)$. Each cell that is not selected is replaced by a new cell with
769 genotype taken at random (uniformly and independently) from the pool of cells that were selected.
770 For example, for $t = 0.8$ and $b = 0$ the top 20 % of fittest cells are selected for the next iteration. The
771 case $t = 1.0$ and $b = 0.2$ represents a selection-free scenario, with a random 20 % of cells surviving
772 each generation. All simulations were carried out using a GPU-based DE package published in [10].

773 **S2 Simulating directed evolution**

774 For simulating DE experiments, the selection and mutagenesis procedures are repeatedly applied
775 to a starting population $\mathbf{p}_0$. The selection function is parameterised by the base chance b , while the
776 rank threshold t is fixed via $t = (1 - f)/(1 - b)$. The fraction of selected cells f is set to $f = 0.2$, so
777 that at each iteration, approximately 20 % of the population is selected for the following iteration.
778 The value of b is determined using the decay rate estimate ρ^{fit} . Before starting DE, the population
779 $\mathbf{p}_0$ is repeatedly mutagenised without selection for M iterations. The resulting decay curve is used
780 to estimate ρ^{fit} , which is in turn used to select b from a pre-computed map that connects ρ_{NK}

781 grid values to optimal selection parameters. This map is pre-computed according to the work
 782 presented in [10]. Next to the parameter b , the work from [10] also incorporates a population
 783 splitting parameter s that subdivides the population p_0 into s smaller populations, resulting in s
 784 smaller DE experiments running in parallel. This is illustrated in Fig 1 (bottom right).

785 The map linking NK ruggedness to the optimal parameter set (b_ρ, s_ρ) is computed using NK land-
 786 scapes, whose grid values are $\rho_{NK} = (K + 1)/N$ via Eq (S2). For fixed N and A , we generate a
 787 number of random NK landscapes for $K = 1, \dots, N - 1$. For each K , we run DE experiments for
 788 different starting populations p_0 and different parameter sets (b, s) , and store the best performing
 789 parameters for each K , defining a map from (N, K) to (b, s) . Given N and the fitted estimate ρ^{fit} ,
 790 the parameters are chosen from the map for which ρ_{NK} is closest to ρ^{fit} . An example is shown in
 791 Fig 1 (middle) for $N = 4$, $A = 20$, and $K \in \{1, 2, 3\}$ ($\rho_{NK} \in \{0.5, 0.75, 1\}$). Note that while ρ^{fit} can take
 792 on continuous positive values, ρ_{NK} can only take on discrete values for fixed N . Alternatively, the
 793 generated maps can be averaged across different values of N .

794 S3 Other Ruggedness Metrics

795 Various measures have been proposed to quantify landscape ruggedness, most of which require
 796 both genotypic and phenotypic information. For our analysis, we compare the analytical decay-rate
 797 measure ρ_2 with five other measures, summarised below. Most metrics were originally defined
 798 on an allele set of size 2 ($|\mathcal{A}| = 2$), and are extended here to larger sets ($|\mathcal{A}| > 2$). For a more
 799 comprehensive discussion of ruggedness measures, we refer the reader to [8]. We also provide
 800 the code used to calculate these metrics at <https://github.com/jessmjames/SLIDE>.

801 Roughness to slope ratio (r/s):

802 Given a genotype $\mathbf{x} = (\sigma_1, \dots, \sigma_N)^T \in \mathcal{G}$, a landscape $f_{lin} : \mathcal{G} \rightarrow \mathbb{R}$ is *linear* if the contribution of each
 803 locus and allele is additive, i.e. $f_{lin}(\mathbf{x})$ can be written as $f_{lin}(\mathbf{x}) = \sum_{i=1}^N \phi_i(\sigma_i)$, where $\phi_i : \mathcal{A} \rightarrow \mathbb{R}$ is
 804 the fitness contribution of locus i for allele σ_i . The roughness to slope ratio [4] captures how well
 805 a given landscape f can be approximated by a linear one: After finding the best linear fit f_{lin} to f ,
 806 i.e. the $N \times A$ values $\phi_i(\sigma)$, the mean slope is defined as $s \stackrel{\text{def}}{=} \sum_{i=1}^N \sum_{\sigma \in \mathcal{A}} |\phi_i(\sigma)| / (AN)$, the roughness
 807 as $r \stackrel{\text{def}}{=} (\sum_{\mathbf{x} \in \mathcal{G}} (f(\mathbf{x}) - f_{lin}(\mathbf{x}))^2 / A^N)^{1/2}$, and the roughness to slope ratio as r/s . For a linear landscape,
 808 $r/s = 0$, whereas for more rugged landscapes r/s increases.

809 Landscape R^2 measure (R^2):

810 The landscape R^2 measure was originally defined in [8] using landscape Fourier analysis and is di-
 811 rectly related to the measure ρ_2 introduced in this work. Suppose that the landscape is represented
 812 in vector notation as $\mathbf{f} \in \mathbb{R}^{A^N}$ and mapped to the Fourier domain as $\hat{\mathbf{f}} = P^* \mathbf{f}$. Define $b_i \stackrel{\text{def}}{=} \sum_{j \in \mathcal{I}_i} |\hat{f}_j|^2$,
 813 where $\mathcal{I}_i$ refers to the indices of $\hat{\mathbf{f}}$ associated with the i th eigenvalue of the adjacency matrix. The

814 R^2 measure is then defined as $R^2 \triangleq (\sum_{i=2}^N b_i) / (\sum_{i=1}^N b_i) \in [0, 1]$, i.e., the higher-order epistasis fraction.
 815 For a linear landscape, $R^2 = 0$, whereas for a more rugged landscape R^2 increases. The R^2 measure
 816 can be computed from decay curves if the full spectrum is estimated via Eqs (20) or (22). However,
 817 the difficulties in estimating the full spectrum can make the R^2 measure less robust than the fit-
 818 ted decay rate ρ_2^{fit} . We refer to this quantity as “ R^2 ” due to it being equivalent to the Coefficient of
 819 determination.

820 Spectral entropy (H):

821 Similarly to the R^2 measure, the spectral or Shannon entropy H uses the power spectral coeffi-
 822 cients b_i and is defined as $H \triangleq -\sum_{i=1}^N p_i \log p_i / \log N \in [0, 1]$ with $p_i \triangleq b_i / \sum_{j=1}^N b_j$. The entropy is
 823 large when the power is spread uniformly across frequencies, reaching its maximum at 1 when
 824 $b_i = b_j \forall i, j \in \{1, \dots, N\}$, and small if the power is concentrated in few coefficients, reaching its
 825 minimum when $b_j = 0 \forall j \neq i$ for some i .

826 Local epistasis (n_ϵ):

827 Epistasis occurs when the fitness contribution of one of the N loci depends on the allele set at
 828 other loci, or in terms of mutations, when the effect of one mutation depends on the presence of
 829 others. Simple sign epistasis occurs when a mutation is beneficial in the presence of one set of
 830 mutations but disadvantageous in another. For example, consider a landscape with $N = 2$ and
 831 $\mathcal{A} = \{0, 1\}$, so that $\mathbf{x} = (\sigma_1, \sigma_2)^T$ with $\sigma_i \in \mathcal{A}$. Simple sign epistasis occurs if changing σ_1 from 0 to 1
 832 has a positive effect on fitness when $\sigma_2 = 0$ and a negative effect when $\sigma_2 = 1$. Here, we measure
 833 local epistasis as the number of genotype pairs with a Hamming distance 2 that exhibit simple sign
 834 epistasis.

835 Distance to local maxima ($d_{\max}$):

836 A local maximum is defined as a genotype with higher fitness than all its neighbours. The metric $d_{\max}$
 837 measures [6] the Hamming distance to the closest local maximum, averaged over all genotypes. A
 838 large $d_{\max}$ indicates less local maxima, hence a smoother landscape.

839 Number of direct paths to global maximum ($n_{\max}$):

840 Given a global fitness maximum at genotype $\mathbf{x}^*$, this metric [7] counts the number of fitness-
 841 increasing direct (length N) paths $n_{\max}$ from the antipodal genotype $\bar{\mathbf{x}}^*$, to $\mathbf{x}^*$. The antipodal geno-
 842 type $\bar{\mathbf{x}}^*$ is the genotype where, for each locus, the corresponding alleles in $\bar{\mathbf{x}}^*$ and $\mathbf{x}^*$ are different.
 843 For $|\mathcal{A}| > 2$, there are multiple antipodal genotypes, so we define $n_{\max}$ to be mean over all $(A - 1)^N$
 844 possible $\bar{\mathbf{x}}^*$. A small $n_{\max}$ indicates a more rugged landscape.

845 S4 Special cases for decay rate metric

846 House-of-cards landscapes:

847 For zero-mean house-of-cards landscapes, the fitness values are assumed to be random, independent, and normally distributed as $f_i \sim \mathcal{N}(0, 1)$. Due to the orthonormality of P , this results in all
848 $\hat{f}_i$ being independent of one another, and $\hat{f}_i \sim \mathcal{N}(0, 1)$, i.e. the Fourier representation inherits the
849 same distribution. As a result, the terms b_i in Eq (7b) are chi-squared distributed with mean equal to
850 the multiplicity of λ_i . The eigenvalue multiplicity $\binom{N}{i}(A-1)^i$ is maximised (The function $\binom{N}{i}(A-1)^i/A^N$
851 can be interpreted as a binomial distribution, in which case $|i - i_c^*|/N \leq 3\sqrt{A-1}/(\sqrt{N}A)$ with
852 $i_c^* = N(A-1)/A$ contains $\sim 99.7\%$ of its weight). for $i^* = \text{round}(N(A-1)/A)$, so that $|b_{i^*}| \gg |b_i|$ for
853 $i \neq i^*$. Substituting the eigenvalue $\lambda_{i^*} = A i^*$ yields the decay rate for house-of-cards landscapes:

$$\rho_{HoC} = \frac{1}{d} \times \lambda_{i^*} = \frac{1}{N(A-1)} \times A \frac{N(A-1)}{A} = 1. \quad (S1)$$

855 Eq (S1) could also be derived differently. Since every fitness value is independent, with an average
856 fitness of 0, every time a cell mutates its new fitness is on average 0. Hence the fitness curve
857 measures how many non-mutated cells are in the population, which is equal to $e^{-\mu}$.

858 NK landscapes:

859 This analysis can be extended to NK landscapes, which can be viewed as a sum of N smaller house-
860 of-cards landscapes of (effective) dimensionality $K+1$ [37]. For each of the smaller landscapes, the
861 eigenvalue multiplicity is maximised for $i^* = \text{round}((K+1)(A-1)/A)$, which substituted in Eq (13)
862 yields

$$\rho \approx \rho_2 \approx \frac{1}{N(A-1)} \times A \frac{(K+1)(A-1)}{A} = \frac{K+1}{N} \stackrel{\text{def}}{=} \rho_{NK}. \quad (S2)$$

863 Since for NK landscapes the parameter $K \leq N-1$ measures the number of epistatic interactions
864 and therefore ruggedness, Eq (S2) shows that the decay rate definition of ruggedness is closely
865 related to the NK notion of ruggedness. Specifically, the analytical decay-rate metric satisfies $\rho \approx$
866 $\rho_2 \approx \rho_{NK}$ for these landscapes.

867 S5 Empirical landscapes and ruggedness

868 To interpret the quantitative predictions of each ruggedness metric it is necessary to consider
869 two factors: the intrinsic properties of the protein and the assay used to generate fitness scores.
870 While we speculate here on differences between landscapes and potential causes for their relative
871 ruggedness, we caution that such interpretations require validation through deeper analysis and
872 complementary measurements to disentangle biological from assay-based contributions.

873 TEV:

874 The data for the TEV landscape was downloaded from [24], and converted into a $20 \times 20 \times 20 \times 20$
875 `numpy` array. The TEV protease (the most rugged in all metrics) recognises a highly specific sub-
876 strate sequence, and its activity depends acutely on maintenance of its catalytic triad [41], as well
877 as substrate-binding pocket and global fold. Catalytic triads in particular are among the least
878 mutation-tolerant motifs in biology - even single substitutions can abolish function - which may
879 contribute to the highly rugged landscape observed. The assay used to collect this data was based
880 on DNA recording with a base editor [24]. This data is inherently noisy due to the added layer
881 of biological complexity, which is likely to contribute significantly to the ruggedness of this land-
882 scape. However, of note is that this assay is much more linearly proportional to activity than the
883 growth-based assays used to measure TrpB and ParD3.

884 ParD3:

885 The data for the ParD3 landscape was downloaded from [22], and converted into a $20 \times 20 \times 20$
886 `numpy` array. ParD3 is an antitoxin in which the C-terminal region is intrinsically disordered in the
887 absence of its toxin partner. Such disordered regions typically rely on general physicochemical
888 properties rather than rigid 3D structures, allowing greater mutational tolerance and producing a
889 smoother landscape [42]. From the assay perspective, this fitness landscape was mapped using a
890 growth competition assay [22]. Such an assay is inherently less noisy than that used to evaluate
891 TEV, due to the fact that the measurement is population level, and the exponential nature of growth
892 also acts to amplify signal at the upper end of fitness. Similarly, growth-based assays of this kind
893 may exhibit “saturation” in the sense that ParD3 variants with toxin binding ability above some
894 threshold at which they can sequester the vast majority of toxin in a cell can thus yield similar
895 (growth-based) fitness scores, even if *in practice* their binding affinity (if it were directly measured)
896 differs significantly. This factor, together with the noise-smoothing properties of this assay, may
897 also contribute to the smoothness of the ParD3 landscape.

898 TrpB:

899 The data for the TrpB landscape was downloaded from [23], and converted into a $20 \times 20 \times 20 \times 20$
900 `numpy` array. Similarly to ParD3, TrpB was evaluated using a growth based-assay [23], allowing us to
901 more confidently compare the two on a biological basis. TrpB is a subunit of tryptophan synthase,
902 which catalyses the synthesis of the amino acid tryptophan. Unlike ParD3, it does not contain
903 any large, intrinsically-disordered regions, and thus many residues may be coupled to maintain
904 the folded structure and activity, contributing to epistasis and therefore landscape ruggedness.
905 Meanwhile, in the case of TrpB this growth coupling is achieved through Tryptophan synthesis
906 which, following a similar argument to above, may “saturate” in its impact on measured fitness for

907 high performing TrpB variants (i.e. once an adequate quantity of Tryptophan is produced in the
908 cell further increase in enzyme efficiency is not rewarded by this assay).

909 GB1:

910 The data for the GB1 landscape was downloaded from [2], and converted into a $20 \times 20 \times 20 \times$
911 `20 numpy` array. GB1 is an immunoglobulin binding protein, whose structure, similarly to TrpB, is
912 globular. Therefore it must also maintain the amino acid interactions that help support folded
913 structure and activity, and consequently contribute to landscape ruggedness. The fitness values in
914 this case were measured using a high-throughput binding assay [2]. The assay achieved a coverage
915 of approximately 100 reads per variant, helping to reduce the noise of the data and therefore assay-
916 related ruggedness, particularly compared to the TEV example.

917 Given the above arguments, we believe that the ordering of these landscapes by ρ_2 (Fig 4g) is jus-
918 tifiable. The assay used to measure TEV fitness was inherently noisy, alongside the fact that the
919 presence of a catalytic triad in the enzyme could add considerable interaction effects and therefore
920 ruggedness. For this reason, it makes sense that it would be positioned above and not with (as it
921 was by local epistasis, paths to max and distance to local max) the rest of the landscapes - and
922 indeed *all* ruggedness metrics investigated agreed with this conclusion. GB1 and TrpB, both glob-
923 ular proteins that were measured with less noisy assays, have justification to be grouped together
924 (as they were by ρ_2 , as well as roughness to slope and landscape R^2) and similarly to be below TEV.
925 ParD3 was positioned at the bottom by all metrics, which aligns with our expectations given both
926 its intrinsically disordered structure and the properties of the growth-based assay used to define
927 its fitness.

928 Considering again Fig 4g and the metric comparisons in Fig 3e we can see significant qualitative
929 differences between metrics. In particular, from Fig 4g the metrics considered can broadly be clas-
930 sified into three groups: Group 1 (decay rate ρ_2 , roughness to slope, and landscape R^2) place TEV
931 as the most rugged, ParD3 as the least, and then group GB1/TrpB close together at intermediate
932 ruggedness. Group 2 (local epistasis and distance to local max) place ParD3 at the bottom, GB1
933 at intermediate ruggedness, but then group TrpB with TEV at the roughness maximum. Group 3
934 (paths to max) in turn places all of TEV/GB1/TrpB at the scale's maximum ruggedness - perhaps not
935 surprising because this metric (as shown in Fig 3e) saturates very quickly as a landscape becomes
936 more rugged. Considering the biological reasoning above we believe our metric (or more broadly
937 those in Group 1) most aligns with the desirable properties of a ruggedness metric (i.e. as a mea-
938 sure of how sensitive a protein's fitness is to mutation): TrpB with its smoothed growth-based data
939 falls below TEV, as too should GB1. An open question remains whether GB1/TrpB *should* exhibit
940 a similar ruggedness value (as seen for all Group 1 metrics) - this would best be answered with
941 further experimental measurements or augmentation of our comparisons with additional protein

942 landscapes as they become available in the future.

943 **S6 Alternative landscape models**

944 In order to explore the effects of other landscape models on the behaviour of SLIDE, both the
945 Rough Mount Fuji and a stochastic block model were implemented (Fig S2). For each model, the
946 optimal strategy choice across low, mid and high ruggedness landscapes was compared. The op-
947 timum strategy for all models tracks from low base chance and low splitting, to high base chance
948 and high splitting as ruggedness increases, validating SLIDE's general transferability across differ-
949 ent landscape models. In addition, it is confirmed that the expected trend in fitness decay curves
950 holds across the different models, with higher ruggedness landscapes exhibiting higher rates of
951 fitness decay in all cases. It can therefore be assumed that SLIDE is not over-fitted to the NK case,
952 and general trends should hold across other landscape models and empirical landscapes.

953 **S7 Alternative mutation models**

954 While this work generally assumes an unbiased mutation spectrum, the assumption that transi-
955 tions between all 20 amino acids are equally probable is biologically unrealistic. For the simula-
956 tions from Section Biased Mutation Spectra, the mutation model is extended. First, the model
957 is extended with codon effects, mapping each three nucleotides to a codon encoding a specific
958 amino acid or a start/stop signal. To incorporate this, a landscape of N amino acids is represented
959 by $3N$ nucleotides. We then use codon mapping to translate these sequences back into the amino
960 acid chain, which is subsequently evaluated using the chosen empirical or synthetic landscape. A
961 critical detail in this process is the treatment of start and stop codons. For this study, we assume
962 a protein becomes completely non-functional if a start or stop codon is inserted anywhere within
963 its sequence, and assigned it the minimal observed fitness. Notably, none of this directly alters the
964 mutation model. Instead, it creates a class of landscapes with an inherent constraint: nucleotide
965 configurations that encode the same protein must share identical fitness values. Because our
966 analysis makes no prior assumptions regarding the fitness values within a landscape, it remains
967 applicable to these models.

968 To account for non-uniform mutation, we use codon mutation likelihoods from [38]. As an example,
969 we symmetrize the *H. sapiens* mutation model by adding the nucleotide transition matrix to its
970 transpose. Analytical ρ_2 values are calculated using Eq 16; however, since this formula is not valid
971 for asymmetric models, we perform the calculation after symmetrisation, effectively using Eq 19.
972 As shown in Fig 6, the curve fits for these models do not converge as closely to the theoretical, global
973 ρ_2 values as they do for NK landscapes or landscapes without codon mapping. To investigate this
974 further, we plot the true decay curves against the theoretical curves using analytical ρ_2 values in
975 Fig S4. The figure shows that the fitted ρ_2^{fit} values align much more closely with the true decay curves

976 than the theoretical values. This discrepancy arises because the eigenvalues are significantly more
977 dispersed in these landscapes, particularly after codon mapping. Increased eigenvalue dispersion
978 degrades the fit of the analytical ρ_2 , which approximates a weighted sum of exponentials with a
979 single one. As Eq 14 shows, the analytical ρ_2 yields a decay curve that acts as a strict lower bound,
980 a trend visible in the plot. The primary exception is the asymmetric mutation on TrPB, where the
981 asymmetry means the lower-bound proof no longer holds. Finally, as implied by Eq 18, the true
982 decay curves remain a strong approximation for the real curves at small μ .

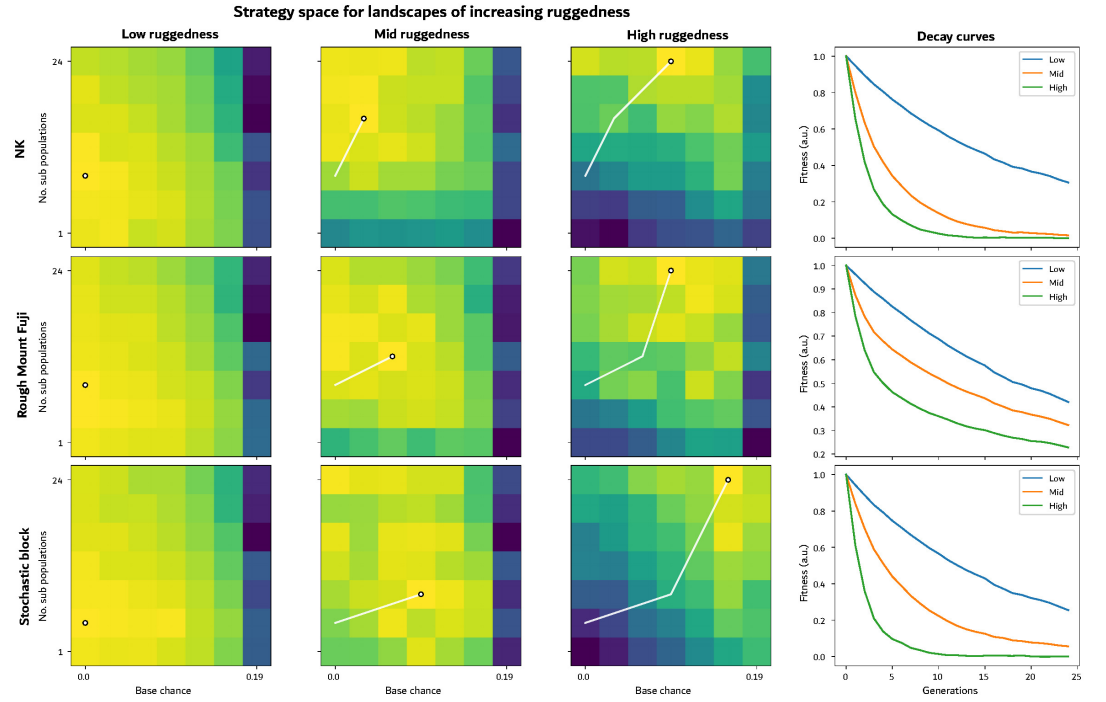


Figure S2. Comparison of strategy space and decay curve trends across different landscape models. Heat maps show strategy spaces across low, mid and high fitness implementations of the NK , Rough Mount Fuji and stochastic block models. White line tracks the optimal strategy as ruggedness increases. Each heat map represents an average across 300 starting points. Line plots on the right hand side display the average fitness decay over the same landscapes represented in the heat maps.

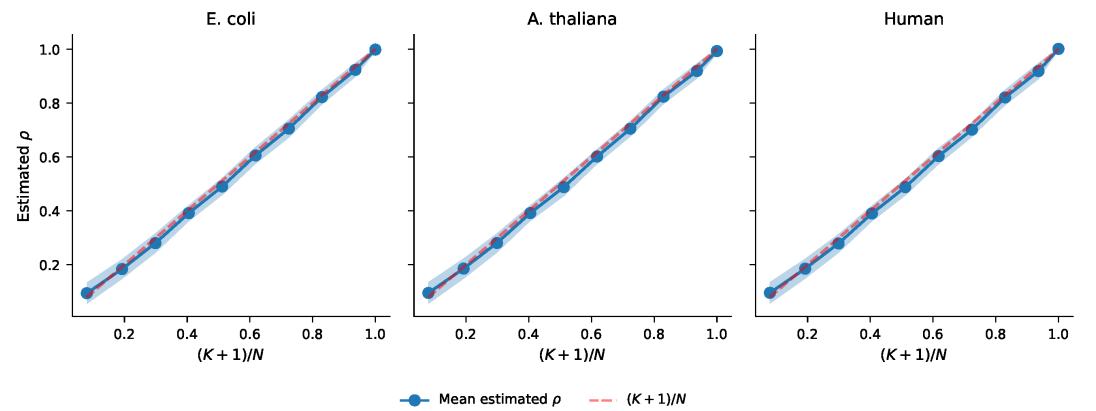


Figure S3. Replication of Fig 3 using biased mutation spectra on NK landscapes. The ρ^{fit} values are estimated across landscapes of increasing ruggedness using 25 generations and a mutation rate $\theta = 0.5$ per generation. The figure shows mean and standard deviation for 300 estimates.

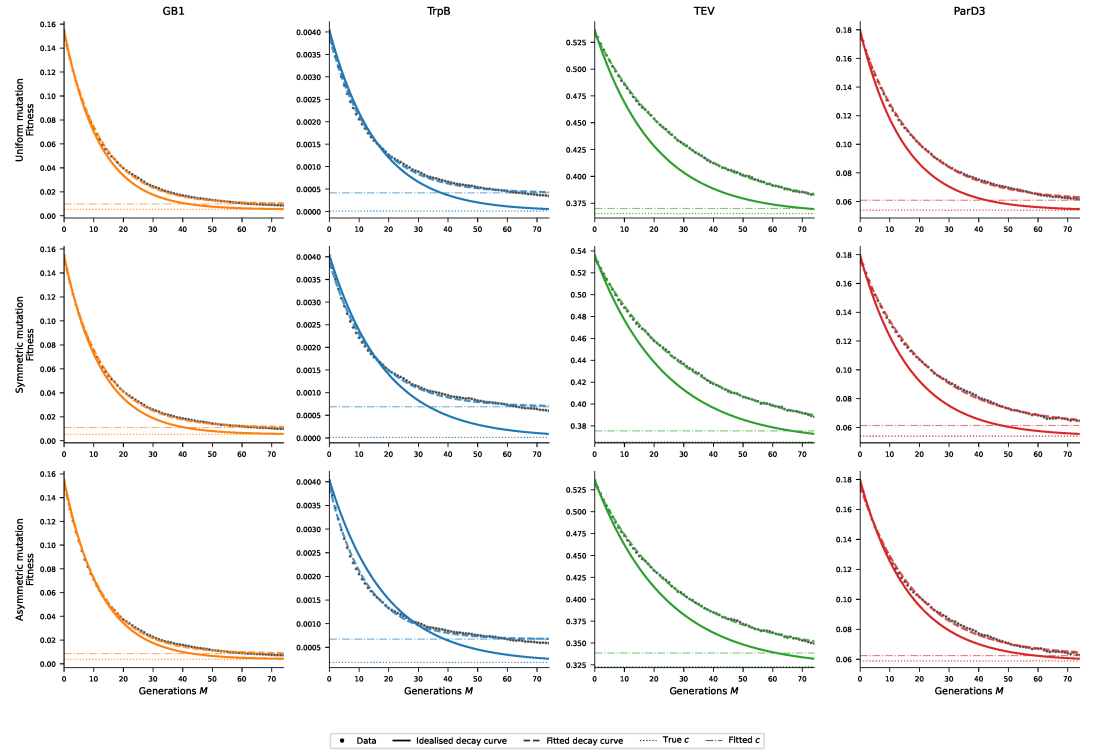


Figure S4. Decay curves on empirical landscapes. The figures show decay curves sampled from empirical landscapes, fitted decay curves, and the idealised decay curves, for unbiased mutations (top), biased symmetric mutations (middle), and biased asymmetric mutations (bottom). To implement biased mutations, all landscapes incorporate a DNA-to-codon mapping. True c represents the true fitness stationary distribution. For the symmetric and uniform mutation models, this is the unweighted landscape average. For the asymmetric mutation model, this is calculated based on the stationary state of the codon transition matrix. Fitted c is the constant term in the exponential fit to the observed data. The mutation rate per generation is $\theta = 0.1/N$.

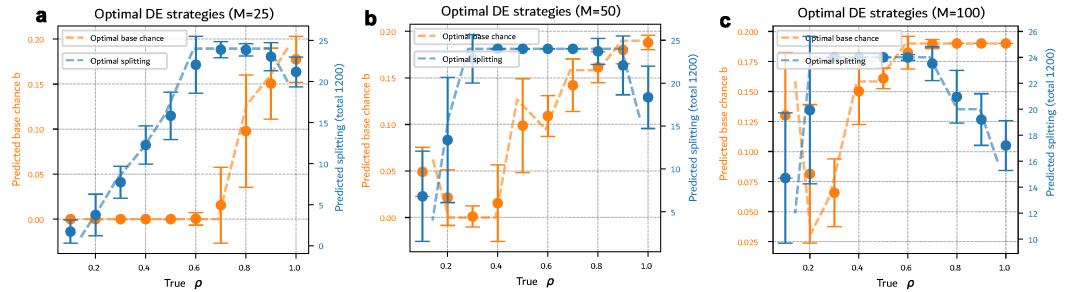


Figure S5. Look up table for NK with higher numbers of DE iterations. Parameters as in Fig 3a but with $M = \text{A: } 25, \text{B: } 50 \text{ and C: } 100$ generations. Error bars represent the accuracy of SLIDE in selecting the true optimal parameters, from 250 simulations.